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Vortex breakdown: a review

O. Lucca-Negro, T. O'Doherty*

Cardiff School of Engineering, Cardiff University, P.O. Box 685, Cardiff, Wales CF24 3TA, UK

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Abstract

This paper reviews the studies undertaken on vortex breakdown over the past 45 years. The paper is structured such that the area is considered in three sections — experimental, numerical and finally theoretical, and provides a 'guide' to the literature and where necessary directs the reader to more indepth reviews in the specific areas. © 2001 Elsevier Science Ltd. All rights reserved.

Keywords: Vortex breakdown; Spiral breakdown; Bubble breakdown; Experimental analysis; Numerical analysis; Theoretical analysis

Contents

1. Introduction . . . 432
   1.1. General introduction . . . 432
   1.2. Recirculation zones . . . 432
   1.3. Precessing vortex core . . . 434
   1.4. Review aims . . . 434
2. Experimental description of vortex breakdown . . . 434
   2.1. Visual observations in tubes . . . 434
   2.2. Types of breakdown . . . 435
      2.2.1. Type 0: axisymmetric mode of vortex breakdown . . . 435
      2.2.2. Type 1 . . . 435
      2.2.3. Type 2: spiral mode . . . 435
      2.2.4. Type 3 . . . 436
      2.2.5. Type 4 . . . 436
      2.2.6. Type 5: double helix mode . . . 436
      2.2.7. Type 6 . . . 436
   2.3. Bubble and spiral forms of breakdown . . . 436
      2.3.1. Axisymmetric or bubble breakdown . . . 436
      2.3.2. Spiral breakdown . . . 437
   2.4. Evolutionary pattern . . . 438
   2.5. Adverse pressure gradient . . . 440
   2.6. Quantitative measurements . . . 441
      2.6.1. Problems encountered . . . 441
      2.6.2. Vortex breakdown flow field . . . 442
      2.6.3. Effects of inlet conditions and geometry . . . 444

* Corresponding author. Tel.: +44-1222-874-797; fax: +44-1222-874-317.
E-mail address: odoherty@cardiff.ac.uk (T. O'Doherty).

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2.6.3.1. Tangential velocity profile and core size . . . 444
2.6.3.2. Axial velocity profile . . . 444
2.6.3.3. Sudden expansion . . . 444
2.6.4. Effect of downstream conditions . . . 444
2.6.5. Structure of vortex breakdown . . . 444
2.7. Overview of the experimental investigations . . . 449
3. Numerical computation of vortex breakdown . . . 449
3.1. Introduction . . . 449
3.2. Quasi-cylindrical approximation . . . 452
3.3. Solutions to the Navier–Stokes equations . . . 452
3.4. Overview of the numerical investigations . . . 462
4. Theories and explanations for vortex breakdown . . . 463
4.1. Introduction . . . 463
4.2. Wave phenomena . . . 463
4.3. Flow stagnation . . . 466
4.4. Stability theory . . . 466
4.5. Combinations of different theories . . . 468
4.6. Other approaches . . . 471
4.7. Bubble and spiral breakdown: same or different phenomena? . . . 475
4.8. Overview of the theoretical investigations . . . 478
References . . . 478

1. Introduction

1.1. General introduction

Swirling flows are observed in natural flows, such as tornadoes and typhoons, and have been widely used, for many years, in technical applications, such as aeronautics, heat exchange, spray drying, separation, combustion, etc. Their importance and complexity have preoccupied research investigations for decades.

In combustion systems, such as in gas turbine engines, diesel engines, industrial burners, and boilers, swirling flows were originally used to improve and control the mixing rate between fuel and oxidant streams in order to achieve flame geometries and heat release rates appropriate to the particular process application [1]. At sufficient degree of swirl, an internal recirculation zone is generated, which allows high rate of heat release, as products of combustion are recirculated and ignite the incoming fuel/oxidant streams. This provides a stable, compact flame [2], with a good performance for difficult carboneous materials and poor quality gases [3].

In aeronautics, leading-edge vortices shed from a delta wing induce a velocity field that results in increased lift and stability of the wing. However, under certain conditions related to the angle of attack of the wing, these vortices can undergo a sudden and drastic change in structure, which can adversely alter the aerodynamic characteristics of the wing. A similar bursting phenomena has been observed for trailing wing-tip vortices, which is, this time, desirable as these vortices represent a hazard to smaller aircraft in areas of dense air traffic [4]. The ability to control these vortical structures is an important and active area of research.

One important feature of swirling flows was put forward by Ranque [5] who discovered the ability of vortex tubes to separate compressed air into hot and cold air streams, located near the periphery and the tube centreline, respectively. This has resulted in a large number of engineering applications utilising this flow behaviour, such as separation of gases of different molecular weight, Ranque–Hilsch vortex tube for a refrigeration scheme, cyclone separation, gas turbine cyclone, combustion chamber, and spray dryers. Despite its simplicity, the mechanism of the Ranque–Hilsh effect has been a matter of long-standing dispute.

Swirl, defined as the ratio of the axial flux of momentum to the axial flux of axial momentum times the equivalent exit radius, has been shown to have large-scale effects on flow fields [6]. Introducing swirl in turbulent jets causes an increase in jet growth, rate of entrainment and rate of decay of the jet. These effects increase with increasing level of swirl.

1.2. Recirculation zones

At low swirl level ($S < 0.6$), there may be significant radial pressure gradients at any axial position due to centrifugal effects, but these do not give rise to more than a slight longitudinal (or axial) pressure gradient which is not strong enough to cause axial recirculation. There is no coupling between the axial and tangential velocity components.

When increasing the swirl, a strong coupling develops between axial and tangential velocity components. A point is reached, when the adverse pressure gradient along the jet

Nomenclature

a	Speed of sound, m/s
A	Constant, m/s
A_t	Cross sectional area of tangential inlets, m^2
B	Constant, $(ms)^{-1}$
C	Complex phase velocity, m/s
C	Relative speed of wave propagation, m/s
D	Diameter, m
F	Frequency, Hz
Fr	Froude number
G	Gravity, m/s^2
G	Velocity profile shape factor
h	Height, m
k	Kinetic energy of turbulence, m^2/s^2
k	Ratio of max swirl velocity to axial velocity
L/D	Bubble length/diameter ratio
m	Azimuthal wavenumber
Ma	Mach number
N	Characteristic parameter for critical regions of a flow
p	Pressure, Pa
Q	Volume flowrate, m^3/s
R	Ratio of the radial to tangential velocity
R	Expansion ratio
r	Radius or radial position, m
r^*	Characteristic viscous length-scale (vortex core radius), m
Re	Reynolds number
Ro	Rossby number
t	Time, s
u	Axial velocity, m/s
$u.\omega$	Helicity density (degree of alignment between velocity and vorticity), m/s^2
v	Radial velocity, m/s
w	Swirl velocity, m/s
x	Axial distance, m
z	Axial coordinate, $(r/\delta)^2$
α	Axial wavenumber
α_0	Ratio of tangential to axial velocity
β_0	Ratio of azimuthal to axial vorticity
δ	Vortex core radius, m
ΔP	Amplitude of the pressure fluctuation, Pa
Δv, Δu	Velocity scale, m/s
ϵ	Dissipation rate of turbulence energy, m^2/s^3
Γ	Local circulation, m^2/s
η	Azimuthal vorticity, s^{-1}
θ	Azimuthal coordinate
ρ	Density, kg/m^3
σ	Radius of a stream surface, m
Ω	Circulation number
Ω	Vortex core rotation rate, s^{-1}
ξ_0	Axial vorticity, s^{-1}

Subscripts

o	Swirl generator chamber diameter
e	Swirl generator exit diameter
1, 2	Refers to condition 1, 2
∞	Upper limit
0	Lower limit

axis cannot be further overcome by the kinetic energy of the fluid particles flowing in the axial direction, and a recirculation flow is set up in the central portion of the jet. The formation of the recirculation flow zone, a form of vortex breakdown, acts as an aerodynamic blockage or the three-dimensional bluff body which serves to stabilize flames [3]. Leibovich [7] describes a vortex breakdown as a change in structure of a vortex initiated by a variation in the characteristic ratio of tangential to axial velocity components.

Although early experimental investigations of vortex breakdown were carried out in flows over delta wings, there was a need for more controlled conditions. Numerous authors investigated the phenomenon in pipe flows [8–13]. Even more constrained conditions, like closed containers with a rotating endwall [14], were used to study the phenomenon.

The method by which swirl is generated has an impact on the form of the vortex core, which in turn, can affect the vortex behaviour. In devices using guide vanes, the viscous core of the vortex originates from the separation of the boundary layer on the centre of the swirl generator. In his guide vane system, Harvey [8] managed to reduce the boundary layer thickness by applying suction near the centre body. He could then vary the ratio of vortex core to pipe diameter. This experience illustrates a characteristic of this type of apparatus: the size of the core decreases as the Reynolds number Re_{tube} (based on the tube diameter) increases. A complicated coupling between the controlling parameters, i.e. flow rate (Re_{tube}) and vane angle (swirl imparted) results from this dependence as an increase of either variable causes an increase in axial vorticity [15]. It would thus seem more reasonable to define the Reynolds number in terms of the vortex-core diameter, which represents a characteristic viscous length scale appropriate for swirling flows [15]. However, most of the qualitative results in the literature are related to the Reynolds number based upon the pipe diameter and comparison between the results of various authors is difficult. In tangential entry generators, the vorticity is mainly due to the rollup of the shear layer shed at the tangential tube exit. A contribution due to separation of the end wall boundary layer also exists but is 'probably a smaller effect' [15]. In Escudier's slit-pipe arrangement [16], this contribution was completely eliminated. Also, the vortex generated by guide vane systems and confined by the pipe, is nearly axisymmetric, while in Escudier's

system, slight asymmetric disturbances occur from Taylor–Gortler vortices in the generator section.

A number of workers, such as Harvey [8] and Sarpkaya [10], have shown that the simplest first breakdown of a vortex takes the form of an axisymmetric bubble of circulating fluid. Initially this occurs several exit diameters downstream of the flow exit. The flow then reestablishes itself before finally breaking down into a spiral form, which is again symmetrical about the central axis. Sarpkaya [10] also showed that this first stage of breakdown of a vortex may be helical in nature, forming a double helix. Further work by Faler and Leibovich [17] showed that the vortex breakdown can in fact occur and exhibit a total of six forms, and was primarily dependent on the Reynolds number. In nearly all cases these forms were considered to be of either a bubble form or spiral form. The recirulation zone is therefore analogous to the bubble form.

1.3. Precessing vortex core

There is evidence to show that the vortex breakdown can often be not only asymmetric, but also highly time dependent [18–22]. This is the result of the forced vortex region of the flow becoming unstable and starting to precess about the axis of symmetry thus forming the so called precessing vortex core or PVC [20,21]. Syred and Beér [20,21] found that the PVC lies on the boundary of the reverse flow zone between the zero velocity and zero streamline. Subsequent work has shown that the presence of a PVC distorts the reverse flow zone into a 'kidney shape' which can lag behind the precessional centre of the vortex by upto 180° [23]. Lucca-Negro [24] showed that the reverse flow zone finally becomes shaped by the PVC forming a helical bubble.

The formation of a vortex breakdown may excite and/or couple with acoustic modes of a system creating pressure oscillations which may alter the fluid dynamic behaviour of the system [25,26]. Syred and Beér [21] highlighted, and Kurosaka [27] confirmed, the presence of a precessing vortex core (PVC) in a Ranque–Hilsh tube and concluded that this phenomenon must contribute substantially to the energy separation process. Similarly, the existence of PVC has also been observed within combustors and dust separation/classification systems. While studies indicate that there maybe a potential benefit in its presence in combustion systems, this has been found to be detrimental in cyclone dust separators [22].

Vonnegut [18] was probably the first to report, without any attempt to explain it, on a time-dependent periodic motion appearing in a vortex whistle: he detected a tone whose frequency was almost directly proportional to the flow rate, for both air and water. In 1957, Peckham and Atkinson [111] observed the breakdown of leading-edge vortices above a delta wing. Lambourne and Bryer [28] also observed a spiral structure on a delta wing and also found that the frequency increased with increasing flow rate. These first observations launched many investigations and reflections.

Despite the extensive experimental, numerical and theoretical research undertaken over several decades, the phenomenon of vortex breakdown remains controversial and no generally accepted explanation has emerged. Comprehensive reviews covering the experimental and theoretical aspects and presenting the progress made in understanding the phenomenon and predicting its occurrence have already been given by Hall [29], Leibovich [7,15], Stuart [30], Escudier [16] and Althaus et al. [31].

1.4. Review aims

The aims of this review are to present a recent 'guide' to the literature, and to emphasise and summarise the significant features of vortex breakdown, the different explanations and also the criticisms or remarks emerging. With the reviews mentioned above as background, it is not necessary to elaborate in great depth on all the contributions, except maybe for the more recent ones, but rather to summarise the information and highlight the salient features.

Also, due to the great number of publications, an extensive review covering all aspects of vortex breakdown cannot be expected. This work mainly addresses vortex breakdown in tubes; other facets, like the breakdown phenomenon over delta wings and in closed containers with a rotating endwall, are mentioned. More detail and wider information can be found in the literature [16,28,31–33].

This paper considers the studies undertaken into vortex breakdown under three main sections, each of them considering one aspect of the phenomenon: experimental, numerical and finally theoretical. This layout is an attempt to structure, as much as possible, all the information contained in the numerous publications on the subject. Obviously, there are many connections between the different parts and an effort has been made to highlight these as often as possible.

2. Experimental description of vortex breakdown

2.1. Visual observations in tubes

Harvey [8] performed experiments on swirling flows within a tube. By varying the swirl level, he found that the breakdown was an intermediate stage between weakly swirling flows, which exhibit no flow reversal, and rapidly swirling flows with columnar flow reversals. The flow reversal was considered as indicative of a critical phenomenon. He described the breakdown as *giving the impression that an imaginary spherical body has been placed on the axis of the vortex, around which the fluid is obliged to flow. In practice, the imaginary object is a bubble which takes the form of a slightly elongated sphere of stagnant fluid downstream of*

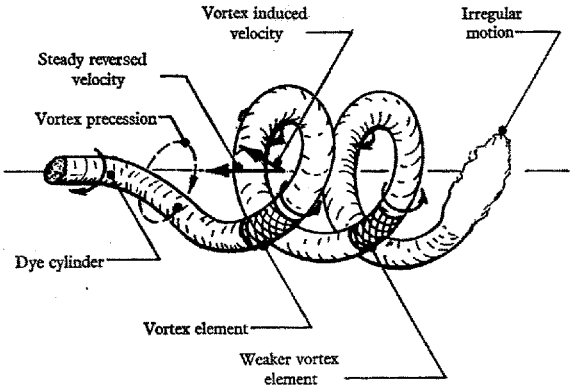


Fig. 1. Sketch of the periodic motion in the vortex whistle [19].

which conditions similar to those ahead of it are restored for a short distance until a second breakdown occurs.

Chanaud [19,34] observed the appearance of an oscillatory motion at certain values of Reynolds number R_{tube}. At low values ($R_{tube} < 300$), the flow was laminar and steady. At $R_{tube} \sim 300$, a small region of reversed steady flow appeared, which grew as the R_{tube} was slightly increased. A disturbance caused by the rapid squeezing of one of the four tangential inlets provoked a regular temporally periodic motion, which faded away after a few cycles. With very small increases in R_{tube}, *too small to be measured*, the periodic motion took longer to evanesce, then remained once caused and, eventually, started naturally without any artificial disturbance. At larger values, the amplitude of the periodic motion increased and a flow resembling a spiral was observed (Fig. 1, similar to the unsteady motion above a delta wing witnessed by Lambourne and Bryer [28]. Chanaud [19] described the motion in terms of an oscillator that derives its energy from the hydrodynamic instability of the steady flow and whose frequency was linked to the characteristic angular velocity of this steady flow. He also suggested that the motion *need not bear a fixed relation to the free stagnation point*. Indeed, increasing the flow rate caused the free stagnation point to move upstream, but had only a minor effect on the axial position where disturbances first became visible.

In their experiments, Cassidy and Falvey [9] also observed a flow remaining steady after the breakdown as long as R_{tube} was very low and the appearance of a helical vortex precessing about the tube centreline as the flow was increased. Harvey [8] observed no spiral.

2.2. Types of breakdown

By varying, systematically and independently, the flow rate and the circulation imparted to the fluid (water) in a slightly diverging cylindrical tube, Sarpkaya [10,11] observed three types of vortex breakdown: double helix, spiral, and axisymmetric (bubble). These are among the seven distinct modes of disruption of the vortex core that Faler and Leibovich's (water) flow visualisations [7] revealed, using a liquid dye tracer, as the Reynolds number R_{tube} (through the volume flow rate) and the swirl (through the vane angle) were varied.

2.2.1. Type 0: axisymmetric mode of vortex breakdown

It is characterised by a stagnation point on the swirl axis, followed by an abrupt expansion of the centreline dye filament to form the envelope of a bubble of recirculating fluid (Fig. 2. This envelope is nearly axisymmetric over most of its length, but the interior is dominated by low frequency motions. Following Sarpkaya [10,11] and Faler and Leibovich [17], this could be caused by fluid exchange with the outer flow at the downstream end of the bubble. A 'tail' can be observed at the rear. The bubble is also only quasi-steady in axial location.

2.2.2. Type 1

This form of breakdown did not have the smooth envelope, distinct emptying tail, and clearly evident downstream vortex core displayed by type 0. The nose of the type 1 breakdown appeared to be slightly asymmetric and the envelope was ragged in appearance.

2.2.3. Type 2: spiral mode

The spiral breakdown is characterised by a rapid deceleration of the dye filament marking the swirl axis, which causes stagnation, and by an abrupt kink, followed by a corkscrew-shaped twisting of the dye (Fig. 3. This latter persists for one or two turns before breaking up into

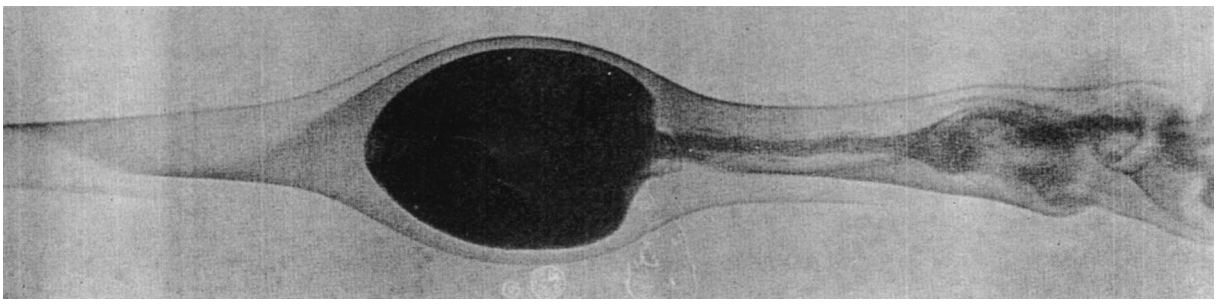


Fig. 2. Bubble type breakdown (from Ref. [10]).

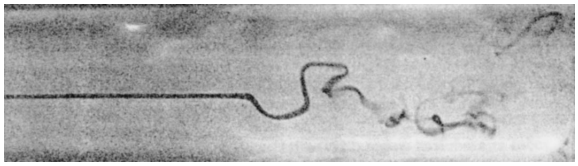


Fig. 3. Spiral type breakdown (from Ref. [7]).

large-scale turbulence. A strong periodicity of the flow pattern is also noticed with the frequency related to the swirl level: an increase in either R_{tube} (with the circulation number Ω constant) or Ω (with R_{tube} constant) is observed to increase the frequency of rotation of the spiral form around the tube axis.

2.2.4. Type 3

This type has characteristics of both the type 4 (flattened bubble) and the type 2 (spiral).

2.2.5. Type 4

The sheared, deflected filament of the type 6 disturbance abruptly began to roll up, back toward the initial point of deflection, into a tight spiral. As this occurred, the filament downstream totally disappeared, indicating that all of the dye reaching the disturbance was being recirculated. After a short time, dye began to exit from this recirculation zone and travel downstream. The recirculation zone was nearly three times as wide as it was thick.

2.2.6. Type 5: double helix mode

A dye filament introduced on the vortex axis is decelerated and expands into a slightly curved triangular sheet. Each half of the sheet is wrapped around the other to form a double helix (Fig. 4). The double helix evolved, in most cases, directly from the type 6 form, described below.

2.2.7. Type 6

The central filament moved gently, but distinctly, off axis at a nearly constant azimuthal location. Only after it had moved off the axis a significant distance did the filament seem to acquire the spiralling sense of the base flow.

Usually, breakdown types 3–6 are less commonly observed than the others and occur only for low Reynolds numbers. At higher Reynolds numbers, the only characteristic geometric forms are the bubble and the spiral [15]. Similarly, above delta wings, only the bubble and spiral forms have been noted, depending on the angle of attack and the wing aspect ratio [35,36]. In a flowfield further constrained than in a tube, for example in the cylindrical container with a rotating endwall investigated by Escudier [14], only the bubble form has been observed. Brown and Lopez [32] suggested that the absence of the spiral form of breakdown in these geometries (at least at the values of height to radius ratio reported) was due to the physical downstream boundary condition which fix the vortex core to the centre ($w = 0$ at $r = 0$).

2.3. Bubble and spiral forms of breakdown

Some more observations made on the two predominant forms are now examined.

2.3.1. Axisymmetric or bubble breakdown

As mentioned before, the bubble continuously exchanges fluid with its environment. Sarpkaya [11] showed through motion pictures the existence of a toroidal vortex ring at the downstream half of the bubble, whose axis gyrates at a regular frequency about the axis of the bubble. Sarpkaya explained the fluid exchange by a simultaneous filling and emptying process through this vortex ring, possibly due to pressure instabilities in the wake of the bubble. The bubble was filled from the side nearer the downstream portion of the ring and emptied from the side nearer the upstream portion of the ring. Faler and Leibovich [17] suggested a similar but opposite process, the filling taking place at the farthest upstream point and the emptying occurring near the farthest downstream point.

The vortex core appears greatly expanded and weaker in the recovery region downstream of the bubble and an axial

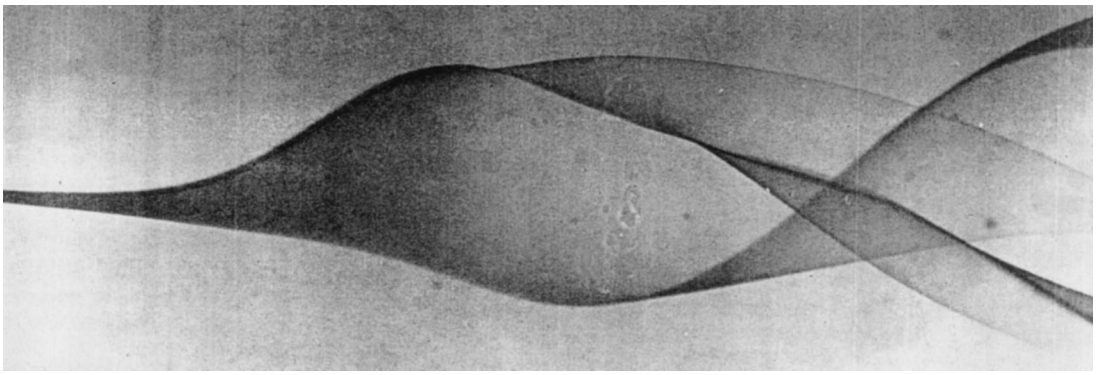


Fig. 4. Double helix breakdown (from Ref. [10]).

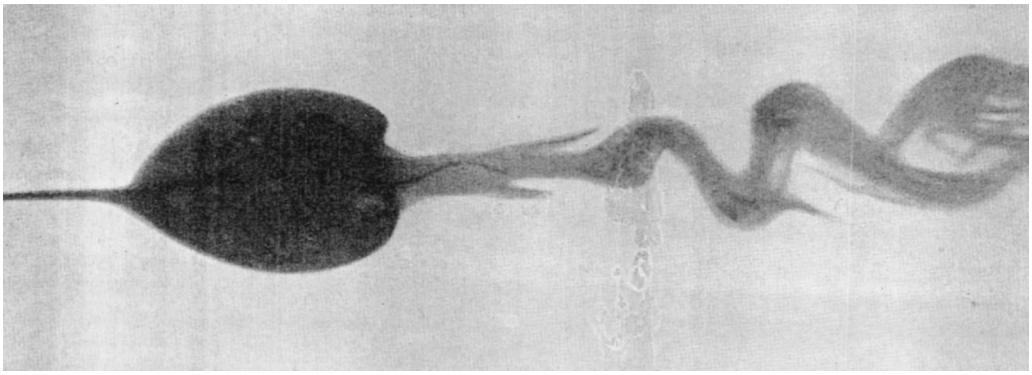


Fig. 5. Bubble breakdown followed by a spiralling tail (from Ref. [10]).

velocity defect is observed, similar to conventional wakes behind bluff bodies [15,37]. Sometimes, the vortex core in the wake appears deflected before breaking up into turbulent fluctuations and a bubble with a 'spiral tail' structure is formed (Fig. 5). Two tails diametrically opposed have occasionally been observed by Faler and Leibovich [17], only at low Reynolds numbers.

The axisymmetric breakdown may evolve from either a double helix, a spiral, or directly from an axisymmetric swelling of the vortex core, the mode of evolution depending on the particular combination of the Reynolds and circulation numbers [11]. For sufficiently high *Re* and circulation numbers, the axisymmetric breakdown bubble came into existence as a result of a perfectly axisymmetric swelling of the vortex-core filament. Escudier [16] presented a series of photographs describing this evolution (Fig. 6).

Starting from a situation in which a spiral breakdown existed far downstream Sarpkaya [11] observed, for an increase in swirl, the appearance of up to two or three axisymmetric bubbles. For further increases in swirl the spiral formed at the downstream end of the bubble train, moved upstream and destroyed the bubbles, with the exception of the first one (Fig. 7). This first bubble, which was closed and perfectly symmetrical during its formative stages, began to draw in fluid from its downstream end and grew in size, while moving simultaneously upstream with its spiralling tail. It finally settled at a point where it would have occurred for the new swirl level.

2.3.2. Spiral breakdown

Sarpkaya [10,11], like Faler and Leibovich [17], observed that the sense of winding of the spiral and the sense of rotation of the whole spiral, as a structure, agreed with that of the base flow. In contrast, Escudier and Zehnder [13] concluded from their experiments that *the sense of the spirals is unmistakably opposite to that of the swirling flow within which they are embedded*, the whole structure still rotating with the outer flow. This is consistent with the spiral observed above delta wings by Lambourne and Bryer [28,38]. Leibovich [15] suggested that the discrepancy might be due to the method of generation of the swirl. As mentioned earlier, flows generated by tangential entry devices [13] are closer to leading-edge vortices than they are to flows generated in vane devices [10,17]. However, Brücker [39], although working with a guide vane system, also observed that the spiral rotates in the same sense as the surrounding fluid, but that the sense of the winding is opposite to it. Another explanation for this disparity between the experimentally observed spiral sense came from the work of Escudier and Zehnder [13] who suggested, for circulation numbers below about six, the existence of a 'prebreakdown' disturbance appearing at lower *Re* than those corresponding to vortex breakdown. In this case, the sense of winding of the spiral was the same as that of the outer flow. Another interesting study carried out by Gursul [40] suggested that the sense of winding depended on the profile of the axial velocity. By exciting the helical mode instability of the vortex breakdown flowfield (the concept of stability of vortex flow is treated further on), he showed that in the case of a jet-like axial velocity profile, the flow is most sensitive to the helical mode disturbances rotating in the opposite direction to the mean swirl, whereas in the case of a wake-like axial velocity profile, the flow is most sensitive to the helical mode disturbances rotating in the same direction as the mean swirl. However, Faler and Leibovich [17], for example, while working with an axial velocity profile of a jet-like type, observed a winding of the spiral in the sense of the outer flow.

Expansion of the vortex core downstream of the breakdown was also observed, though significantly smaller than the one associated with the bubble form [15].

Lambourne and Bryer [28] noticed that although the dye filament had a spiral form, an individual particle, after moving along the straight portion of the filament upstream of the breakdown point, was deflected, while being convected along with the convolution of the dye filament of which it is a part, along a path which hardly turned at all about the central axis (path SPP' in Fig. 8). Hall [29] explained this observation by the fact that the fluid initially close to the axis has little angular momentum.

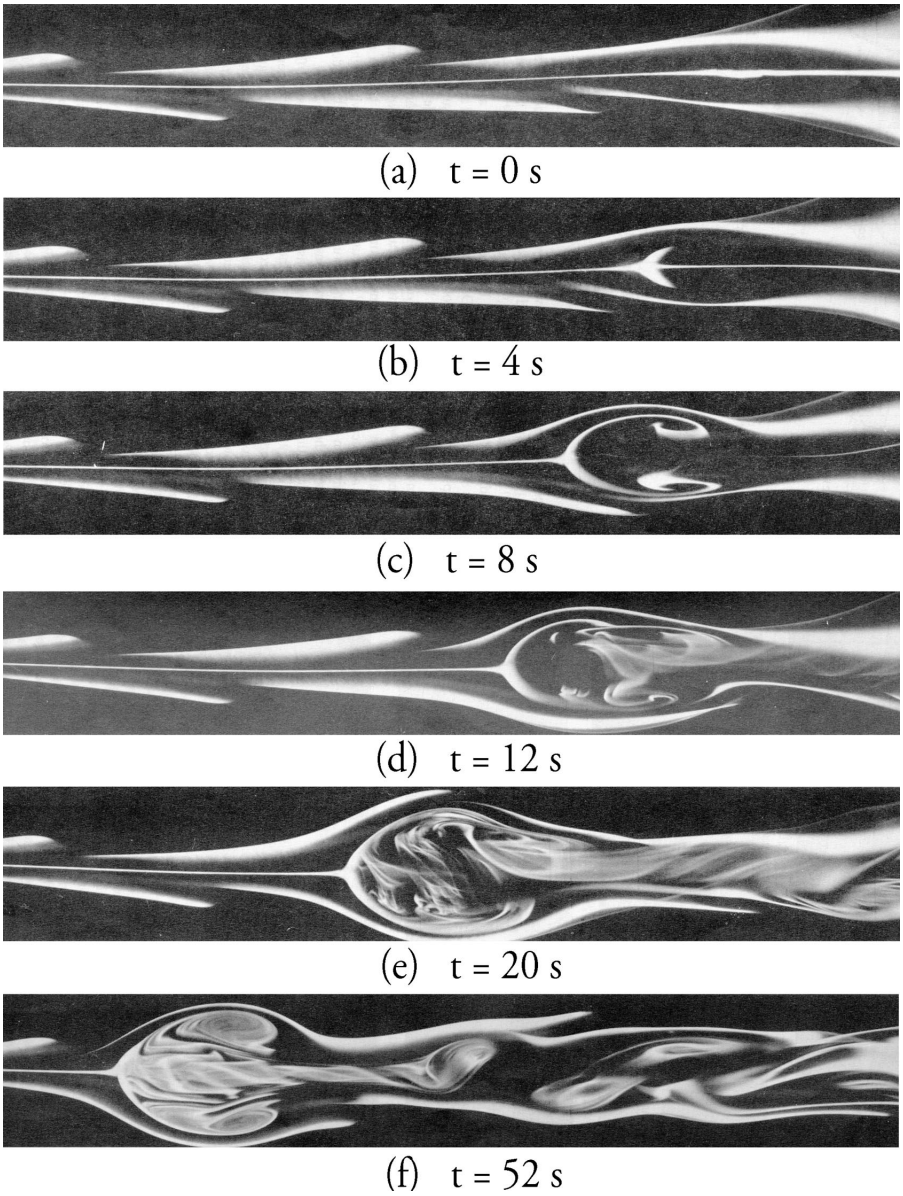


Fig. 6. Formation of an axisymmetric bubble by core swelling (from Ref. [16]).

2.4. Evolutionary pattern

The different forms observed by Sarpkaya [10] and by Faler and Leibovich [17] followed a definite order of progression as the flow parameters were varied, and there was a clear evolutionary pattern. All breakdowns in this hierarchy, including the bubble form, were asymmetric in azimuth. The charts of mean position and mode versus the flow parameters established by these authors are presented in Figs. 9 and 10 (where $\Omega = \pi \Gamma D_c/4Q$). Average Reynolds numbers (based on the throat diameter) varied from less than 1000 to greater than 10 000. The positions are only mean

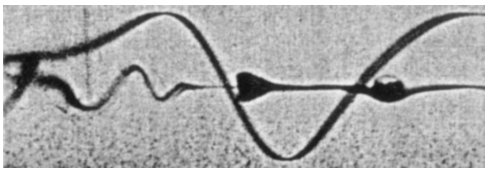


Fig. 7. Multiple breakdown (from Ref. [11]).

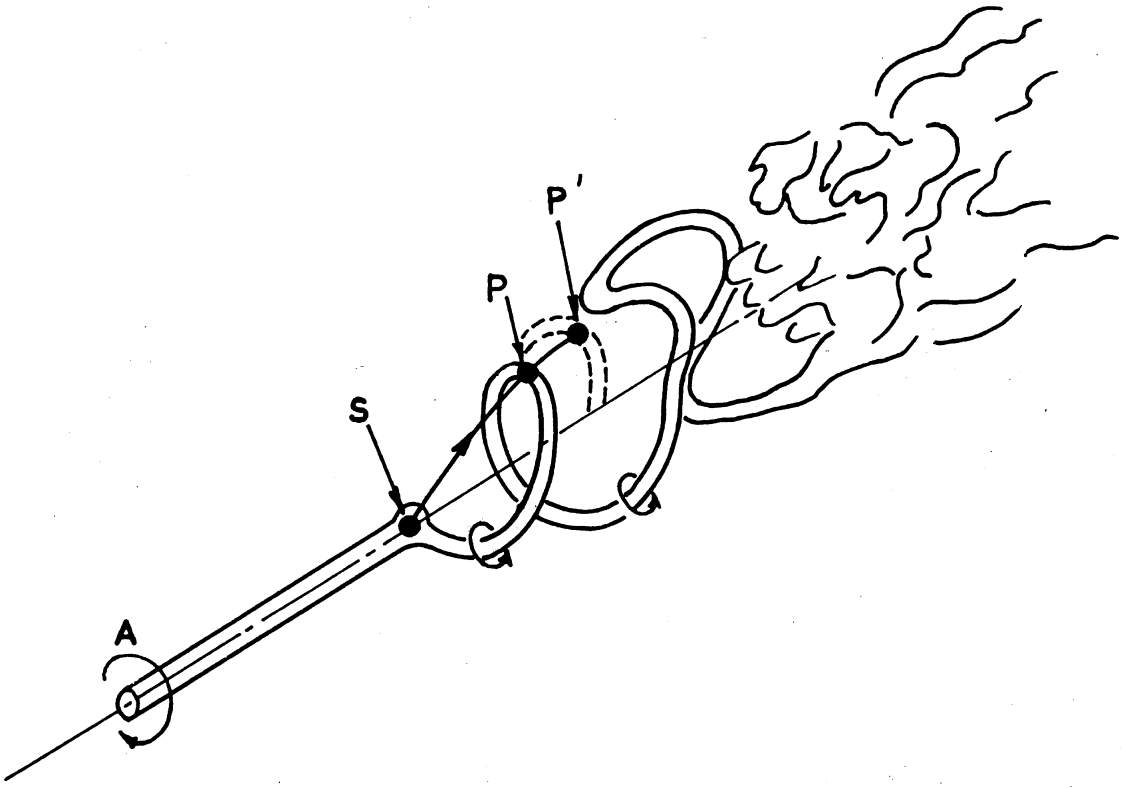


Fig. 8. Behaviour of the axial filament [38].

values as breakdowns exhibit a considerable amount of positional instability, tending to wander back and forth along the tube axis with no noticeable change in external conditions. However, this wandering occurs within specific limits about a mean position which is *reasonably repeatable* for any fixed set of flow conditions [7].

It clearly appears from these figures that, in the progression of forms as the swirl is increased, spiral breakdown *always* occurs before axisymmetric breakdown, which is the final form. It also appears that the bubble's breakdown position is always further upstream than that of the spiral's. An increase in swirl, starting from a low value, causes an upstream movement of the spiral. When the swirl level is sufficiently large, the spiral transforms into a bubble form.

As was also observed by Escudier and Zehnder [13], the vortex breakdown becomes increasingly steady as the circulation number is increased. Further increases in swirl produce no further change in pattern and only moves the breakdown further upstream, until it reaches the upstream boundary of the equipment. According to Harvey [8], if the swirl is increased beyond this level a columnar vortex is formed, with backflow along the entire axis of the tube. An increase in the swirl level tends not only to move the bubble upstream, but also tends to increase the rotational frequency of the tail and to move it closer to the axis. A transformation from two tails to a single tail could be observed when slightly increasing the Reynolds number and/or the swirl [17].

Sarpkaya [10] presented an arresting description of the response of the bubble breakdown to a sudden variation of the swirl: the breakdown first moved a short distance in the direction opposite to that expected and then rapidly toward the final equilibrium position, overshooting it. Finally, the breakdown slowly relaxed backwards to the position required by the new swirl level. A similar observation was made by Faler and Leibovich [17]. This motion illustrates the nonlinear nature of the flow.

While there is a definite progression of forms, there are numerous borderline ranges of flow settings for which two forms (or more) can exist and transform spontaneously into each other. Even a periodic transition between bubble- and spiral-type breakdown has been observed [36]. Sarpkaya [11] called this zone a hysteresis region, as he observed that the initial form, spiral or bubble, was conserved when increasing or decreasing the swirl level respectively. However, Faler and Leibovich [17] contested this classification as there was no evidence in their experiments that the type of breakdown formed depended on how this flow setting was achieved. Faler and Leibovich [12] pointed out that any transformation between the two forms causes an axial movement of the newly established breakdown to the mean axial position appropriate to its type (see Fig. 10). If

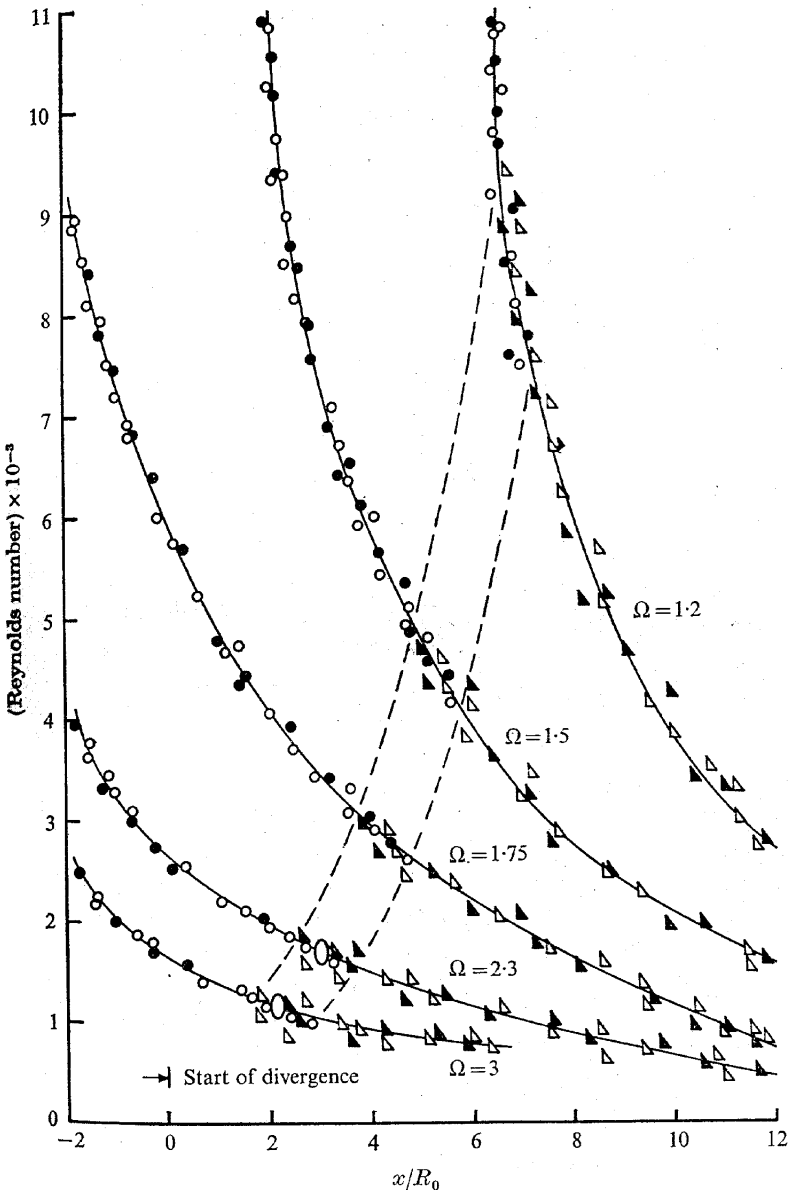


Fig. 9. Vortex breakdown position as a function of Reynolds and circulation numbers [10].

the spiral changes to a bubble, it travels several vortex-core diameters upstream and if the bubble becomes unstable and changes into a spiral, this latter travels downstream again. Sarpkaya [10] did not notice these motions (Fig. 9). Except for these spontaneous transformations, both spiral and axisymmetric vortex breakdowns show high degrees of structural stability.

2.5. Adverse pressure gradient

Working on delta wings, Lowson [41] studied the vortex breakdown position. He noticed that if the length of the wing was doubled, the breakdown's position was at twice the distance from the apex. He stated that the pressure recovery over the wing due to the trailing edge was the primary factor governing the final position of the breakdown. In their experiments, Cassidy and Falvey [9] observed a substantially increased frequency of precession of the helical vortex if the exit of the tube was partially blocked. Similarly, when the flow was decelerated by constricting the downstream end of the tube, Sarpkaya's visualisations [10] showed that the breakdown rapidly moved upstream or in some cases could even be initiated. The common point between these different observations is the existence of a positive or

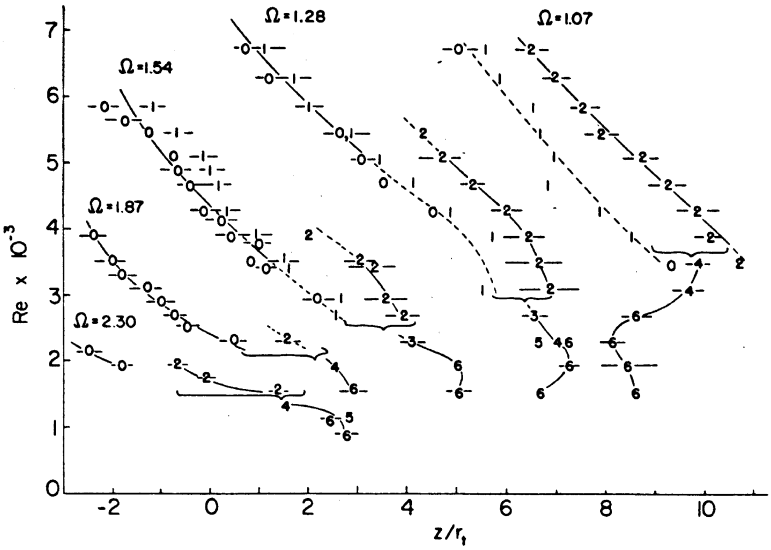


Fig. 10. Vortex breakdown position as a function of Reynolds and circulation numbers [17].

adverse axial pressure gradient. In these examples, this gradient is imposed externally. However, as Hall [42] was the first to demonstrate that axial pressure gradients in swirling flows consist of two components: the imposed external pressure gradient and the contribution due to the swirl. Indeed, pressure gradients imposed at the edge of the vortex core are amplified on the centreline when high swirl velocities are applied. Thus, under certain circumstances, a positive axial pressure gradient can exist even if the imposed pressure gradient is negative. For an inviscid flow, Escudier [16] wrote:

$$\left(\frac{\mathrm{d}p}{\mathrm{d}x}\right)_0 = \left(\frac{\mathrm{d}p}{\mathrm{d}x}\right)_\infty + \rho\int_0^\infty \frac{\Gamma}{r^3}\frac{\partial\Gamma}{\partial r}\frac{v}{u}\,\mathrm{d}r \tag{1}$$

wherein $\mathrm{d}p/\mathrm{d}x|_0$ is the pressure gradient on the axis ($r = 0$), $\mathrm{d}p/dx|_\infty$ is the imposed pressure gradient (at $r = \infty$), and $\Gamma = 2\pi rw$ is the local circulation. A typical tangential velocity profile formed from a swirl generator [43] is a combination of a forced and free vortex distribution and is commonly known as a Rankine vortex. Considering this distribution, the equation becomes:

$$\left(\frac{\mathrm{d}p}{\mathrm{d}x}\right)_0 = \left(\frac{\mathrm{d}p}{\mathrm{d}x}\right)_\infty + \frac{2\rho}{\delta^3}\left(\frac{\Gamma_\infty}{2\pi}\right)^2\frac{\mathrm{d}\delta}{\mathrm{d}x} \tag{2}$$

This equation shows that the swirl contribution is positive if the core grows; its importance increases if the core is smaller, if the circulation is increased, or if core expands more rapidly.

Similarly, Krause [44] showed, from a simple analysis of the momentum equations for a flow upstream of the point of breakdown, that if the core grows in the streamwise direction, i.e. the radial velocity is always positive, breakdown must eventually occur. The analysis assumed an incompressible, steady and axisymmetric flow. He also concluded that the distance between the point of initiation and of breakdown of the vortex is inversely proportional to the square of the maximum azimuthal velocity and directly proportional to the square of the freestream velocity. This means that vortex breakdown can be delayed by decreasing the former and increasing the latter.

Usually, then, as also shown numerically by Narain [45], increases in the swirl level, in the adverse pressure gradient, or in the degree of divergence of the flow tend to move a breakdown further upstream and could transform a spiral type into a bubble. However, Sarpkaya [46] demonstrated that an increase in the degree of divergence beyond a certain limit, for which the boundary layer separates from the diverging wall, does not necessarily cause the breakdown to travel further upstream. This interaction between breakdown and boundary layer separation illustrates well what Hall [29] pointed out, that breakdown is not just governed by a few parameters, but that the *overall* pressure and velocity fields are very important.

2.6. Quantitative measurements

2.6.1. Problems encountered

Harvey [8] succinctly described two serious problems encountered when attempting to take measurements in vortex breakdown flows. One problem arises from the fact that vortex breakdown is highly sensitive to external disturbances, and introduction of probes can dramatically perturb the flow. Cassidy and Falvey [9] observed a rapid upstream motion of the breakdown *if any type of probe were placed in the flow*. The random variations of vortex breakdown about a mean position present another difficulty. Early

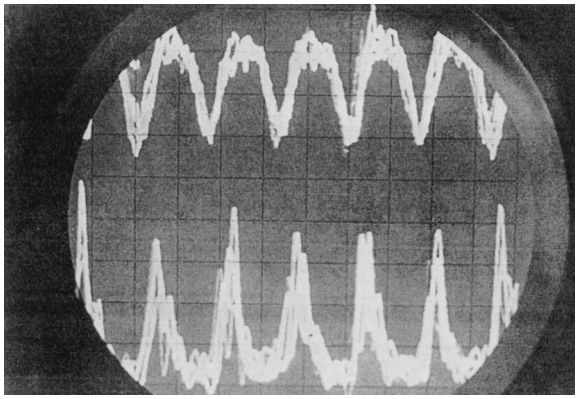


Fig. 11. Velocity and pressure traces [9].

measurements involved different sorts of probes susceptible to altering the character of the flow, and are therefore not very reliable. However, nonintrusive methods were quite rapidly adopted, such as LDA (Laser-Doppler Anemometry), PTV (Particle-Track Velocimetry), etc.

2.6.2. Vortex breakdown flow field

From velocity measurements, a schematic description of vortex breakdown flow fields could be obtained. Leibovich [7] suggested a division into three spatial regimes:

1. The approach flow consists of a concentrated vortex core embedded in a flow that often may be approximated as irrotational. The profiles of axial velocity are jet-like. It is characterised by small axial gradients.
2. The breakdown region follows. It is characterised by rapid changes in the axial direction, with deceleration of the approach flow, formation of a stagnation point on the vortex axis, flow reversal near the axis and eventually restoration of the original direction of the axial flow.
3. A new vortex structure with an expanded core is established downstream of the breakdown zone. Axial-velocity profiles closely resemble a conventional wake behind a solid obstacle, with centreline speeds lower than those outside the vortex core. This region is again characterised by small axial gradients, and the flow is invariably (more) turbulent further downstream.

Time-averaged axial and tangential velocity profiles *upstream* of a bubble and a spiral breakdown were presented by Nakamura and Uchida [47]. These are similar for both types of breakdown except at the cross-section where stagnation occurs: the axial velocity near the wall is higher for the spiral. This corresponds to a smaller adverse pressure gradient.

Cassidy and Falvey [9] observed the signals from a hot-film anemometer and a pressure cell as the angular momentum flux was gradually increased. When the probes were positioned closely together, the velocity and pressure were seen to have the same frequency (Fig. 11). Their experimental results could be correlated in a dimensionless form using:

1. The momentum parameter $\Omega D/\rho Q^2$, proportional to the geometric swirl number $\pi D_o D_e/4A_t$, where Ω is the flux of angular momentum.
2. The frequency parameter fD^3/Q, proportional to the Strouhal number, fD/u.

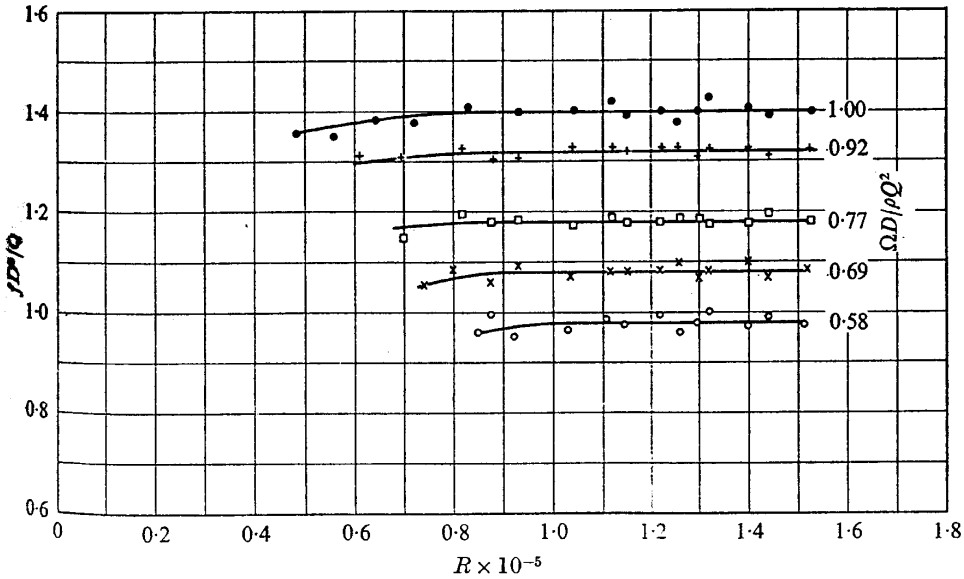


Fig. 12. Frequency parameter vs Reynolds number and momentum parameter [9].

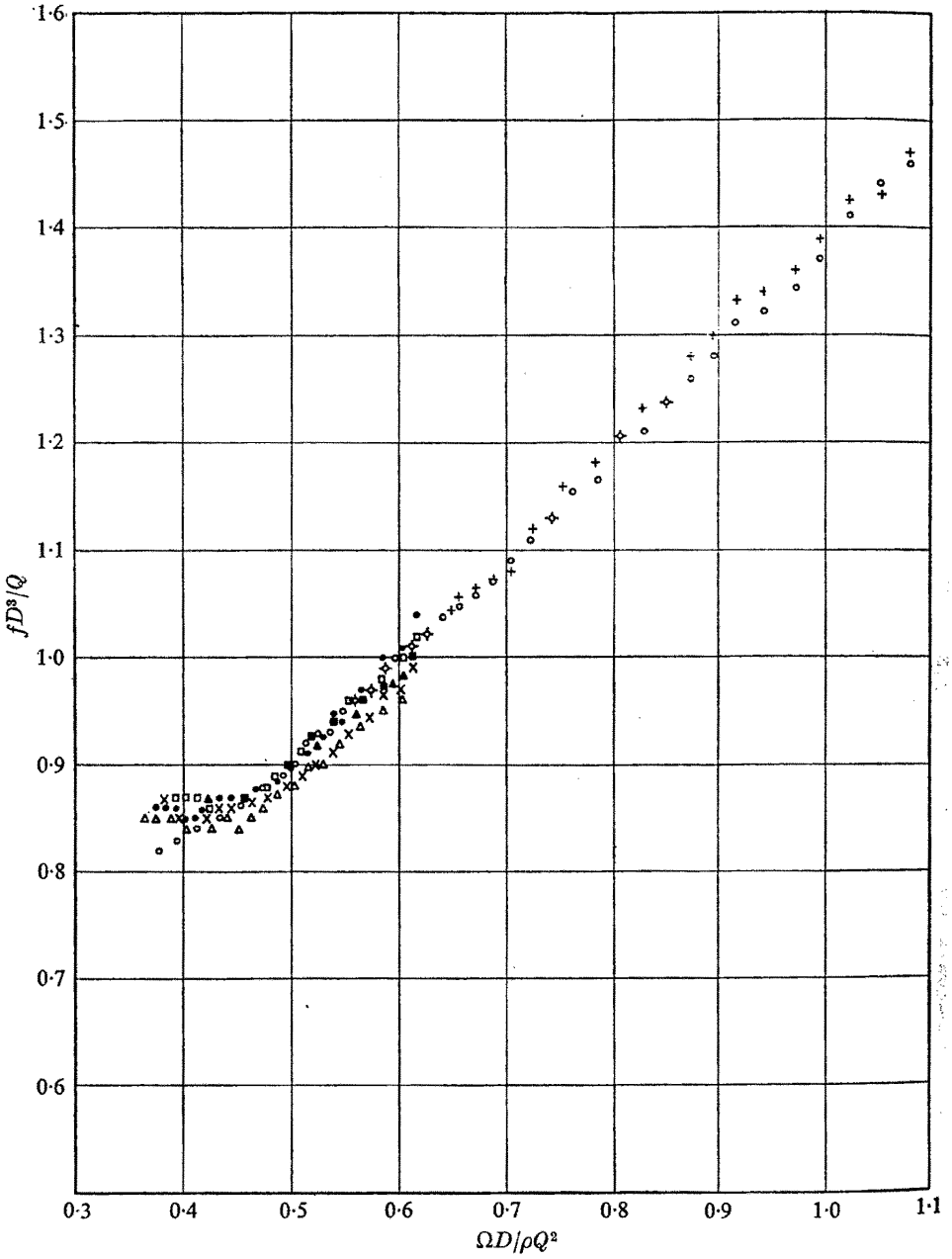


Fig. 13. Frequency parameter vs momentum parameter at high *Re* [9].

3. The pressure parameter $\Delta PD^3/\Omega$, where ΔP represents the amplitude of the pressure fluctuation.

They plotted fD^3/Q as a function of the Reynolds number Re_{tube} (Fig. 12) and concluded that the frequency of the flow is essentially independent of viscosity for large Re_{tube} and tends to become constant for a particular value of $\Omega D/\rho Q^2$.

Values of fD^3/Q plotted against $\Omega D/\rho Q^2$, for the range in which *viscous effects are not important* (Fig. 13), show that frequency increases rapidly with increasing $\Omega D/\rho Q^2$. Similar conclusions were reached for the amplitude of the pressure fluctuation.

As previously seen, increasing Re_{tube} decreases the core radius and increases the axial vorticity. In addition, the Re_{tube} above which the frequency parameter (and also the pressure parameter) becomes constant for a given momentum parameter ($Re \sim 6000$–$10\,000$) is much higher than the Reynolds number marking the transition between laminar

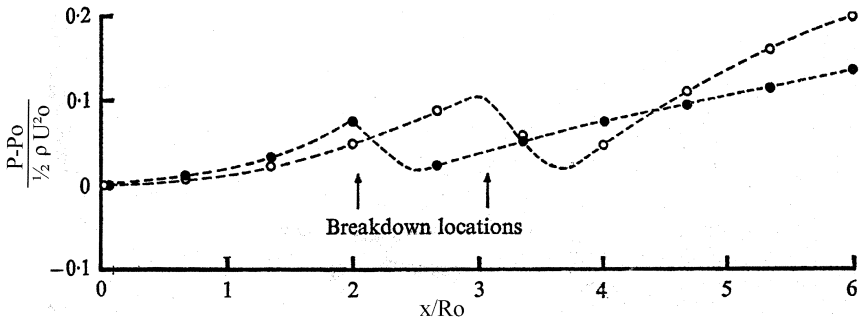


Fig. 14. Evolution of the pressure excess ratio in the streamwise direction [10].

and turbulent states ($Re \sim 2000$–3000). The effect of viscosity mentioned by Cassidy and Falvey might thus rather be an effect of vorticity.

Wall pressure measurements were presented by Sarpkaya [10] in terms of a pressure excess ratio $(p - p_u)/1/2\rho u^2$, where p_u is the pressure far upstream of breakdown and u is the mean axial velocity (Fig. 14). Upstream, the pressure excess ratio was initially zero but increased slowly approaching the breakdown, where it reached a local maximum, decreased rapidly through the breakdown region and then slowly increased to a final constant value in the downstream region. Similar results had been observed by Kirpatrick [48].

2.6.3. Effects of inlet conditions and geometry

Effects of inlet conditions and geometry on the onset of flow reversal have been studied by several authors [49,50].

2.6.3.1. Tangential velocity profile and core size. Experimenting with swirling flow exhausting into a sudden expansion, a condition which is often present in combustion systems, Hallett and Toews [49] concluded that a tangential velocity distribution with a large solid body vortex core leads to a higher 'critical' swirl. 'Critical' is used here in the sense 'minimum required for flow reversal'; it does not refer to Benjamin's definition [51], discussed later in this paper. Farokhi et al. [50] also highlighted the fact that two swirling jets with the same flow rate, Mach number and swirl number, but different initial swirl profile, show different evolutions. Predominantly free-vortex distribution (as opposed to solid-body rotation) causes a smaller vortex core, a larger radial inflow and a larger static pressure deficit in the core. They stated that the integrated swirl effect, reflected in the swirl number, is inadequate in describing the swirling jet behaviour in the near field and that the relative size of the vortex core and the corresponding tangential velocity distribution are the controlling parameters.

2.6.3.2. Axial velocity profile. Hallett and Toews [49] also deduced that a flow exhibiting inlet axial velocity profiles with a maximum on the axis also required a higher level of swirl to produce vortex breakdown, as a higher positive pressure gradient was required to obtained flow reversal.

2.6.3.3. Sudden expansion. The influence of a sudden expansion on vortex breakdown can also be understood in terms of pressure gradient: the vortex core grows rapidly and generates an adverse pressure gradient, which is further enhanced by the pressure recovery due to the general decrease in axial velocity. This combined effect leads to a strong reduction in critical swirl with increasing expansion ratio up to $(R_2/R_1) \sim 2$, for which the pressure recovery reaches a maximum in nonswirling flows before again falling to zero. For higher expansion ratios, as confirmed by the experiments of Beltagui and MacCallum [52] and of Rao et al. [53], the critical swirl does not change a lot.

2.6.4. Effect of downstream conditions

Downstream influence has also been observed by different investigators. Without swirl, an exit contraction can cause a global increase in pressure upstream of the exit and an acceleration of the flow in the vicinity of the exit. In the case of swirling flows, these effects can be further complicated and the entire flowfield can be affected. Escudier et al.'s [54] LDA measurements revealed a dramatic change in the vortex structure as the exit diameter was reduced (Figs. 15 and 16).

They noted that 'flow visualisation indicates the existence of low velocities in the core centre even for small exits, and it is likely that only the resolution of the LDA instruments prevents this from being seen in the axial velocity profiles'. Ref. [112] also showed (Fig. 17) that even a weak contraction can have a significant influence on the shape of the recirculation zone, if the flow is subcritical (i.e. capable of supporting standing waves) between the recirculation zone and the exit. For very strong contractions, the recirculation zone can exhibit an annular shape without flow reversal on the axis. This effect has also been observed by Chao et al. [56].

2.6.5. Structure of vortex breakdown

The first quantitative information about the flow structure

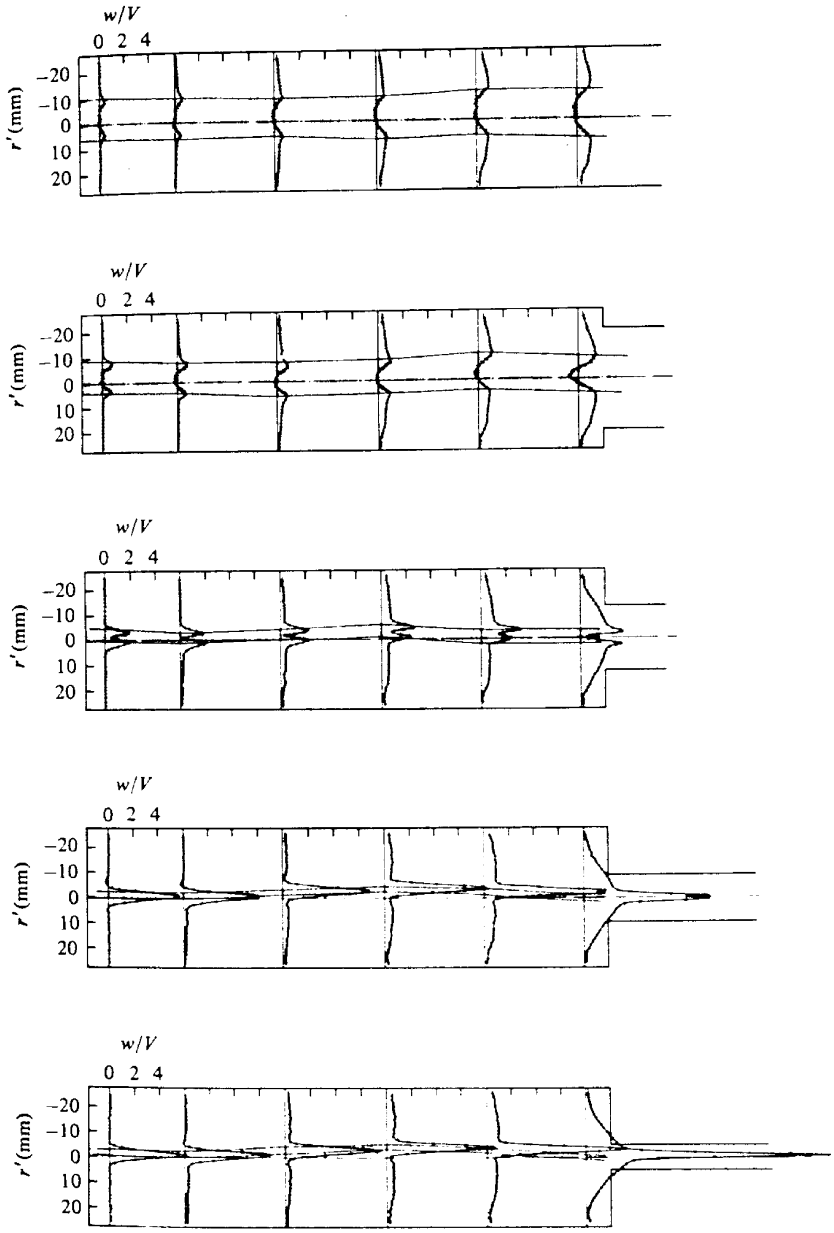


Fig. 15. Effect of an exit contraction on the normalised axial velocity distributions [54].

of vortex breakdown was published by Faler and Leibovich [12] for a bubble type breakdown. The apparatus used was similar in design to Sarpkaya's [10] and the regions were mapped using LDA techniques. These measurements confirmed Sarpkaya's [10,11] observation that the flow in the bubble form of breakdown is asymmetric and periodic in time and azimuth. However, the mean streamlines constructed from time-averaged axial velocity measurements (under the assumption that the time-averaged motion is axisymmetric) revealed a two-celled structure in the interior of the recirculation zone, with a total of four stagnation points on the axis (Fig. 18). In the inner cell, the velocities near the axis are oriented downstream. The two cells were considered to be responsible for the exchange of fluid at the rear of the bubble: *the outer ring fills the bubble*, *whereas the inner counterrotating vortex ring empties it*. Similar results were presented by Bornstein and Escudier [57] through LDA measurements, although these did not suggest a closed bubble with two internal stagnation points. A sketch drawn from these measurements is compared with a photograph in Fig. 19.

Faler and Leibovich [12] observed the velocity

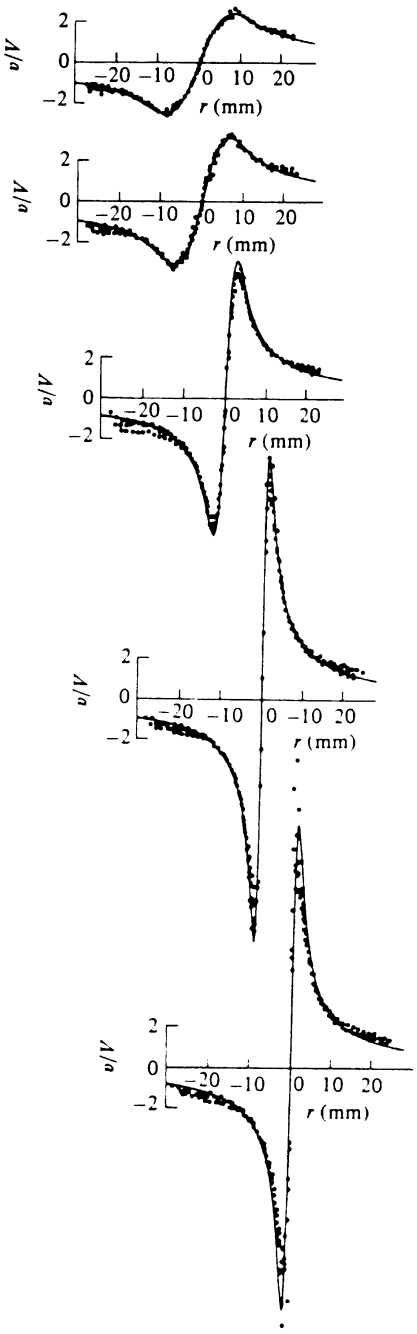


Fig. 16. Effect of an exit contraction on the normalised swirl velocity distributions [54].

fluctuations to vary in nature within the bubble: in the nose region, fluctuations were small and random, with no apparent dominant frequency, while in other regions of the bubble (especially in the downstream half) and in the wake regions, they were strong and highly periodic.

Later, LDV (Laser Doppler Velocimetry) measurements were presented by Uchida et al. [58]. An axisymmetric breakdown followed by a spiral was observed. In the upstream region of the bubble, the axial velocity on the axis settles to a constant, positive, and low value. However, in the downstream half, it abruptly increases, before decreasing again when reaching the second, spiral, breakdown. The schematic diagram of the flow pattern proposed is shown in Fig. 20.

Brücker and Althaus [59] obtained velocity and vorticity fields within the bubble-type breakdown region using PTV (Particle Tracking Velocimetry) combined with laser-sheet scanning. Brücker [39] extended the study to a spiral breakdown. As was first noted by Sarpkaya [10], a stagnation point was detected on the vortical axis, followed by a limited region of reversed axial flow near the vortex centreline. An overall flow reversal was observed inside the bubble near the centreline, which contrasted with measurements from Refs. [12,57,58]. The quasi 3D instantaneous measurement technique also highlighted the exchange of fluid at the downstream open end of the bubble. A *single* tilted ring-like structure, gyrating around the centreline, appears to dominate the interior flow structure: the bubble seemed to be filled from the downstream part of the vortex ring and emptied from its upstream part. These measurements and their interpretation agreed qualitatively with Sarpkaya's observations [10]. Although the vortical flow bulged over the bubble and a stagnation point was generated, the vortex axis seemed to remain at the centreline. Downstream of the stagnation point, the axial vorticity within the bubble was found to be strongly reduced, except around the edges of the bubbles. In the downstream part of the bubble and further downstream, the axial vorticity was again concentrated on the axis and appeared to be intensified, while the characteristic size of the vortex core had decreased. Contrary to this, in the case of a spiral breakdown, the vortex axis was shown to depart from the centreline [39]. The stagnation point also appeared to be located outside of the centreline and rotated around it as the outer flow. This is an important difference compared to bubble breakdown. It was argued that the position of the stagnation point was linked to the slope of the deflected vortex core, via the induced backflow (Fig. 21).

A comparison of the measurements for a bubble and a spiral breakdown indicated a strong similarity between the two forms and Brücker [39] stated that *the slope, winding and diameter of the spiral vortex core determine the different observable forms*. For him, the bubble breakdown appeared as a 'compressed' spiral.

A more detailed discussion concerning the similitudes and differences between bubble and spiral forms of breakdown will come later, when discussing the different explanations to the vortex breakdown phenomenon. However, it is important to underline that Brücker's explanation of the bubble breakdown is not necessarily incompatible with structures presented by Faler and Leibovich [12] or Bornstein and Escudier [57], and shown, respectively, in Figs. 18 and 19. More especially as these different

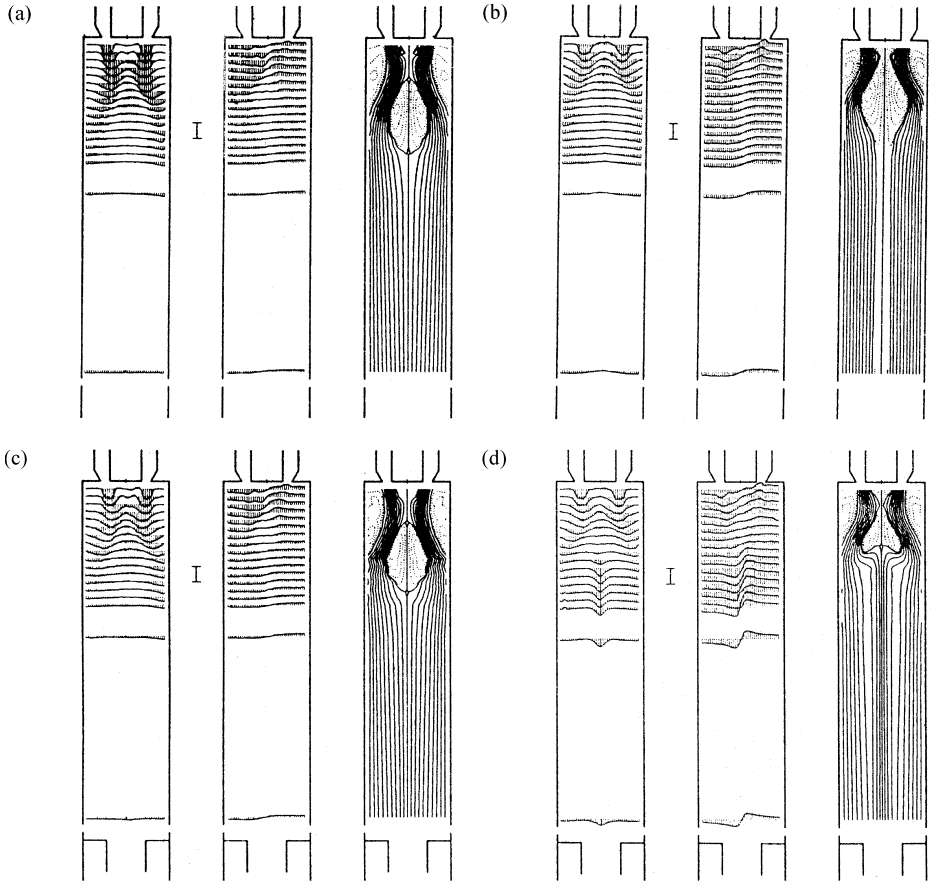


Fig. 17. LDA measurements-contraction 79% exit area ($De/D = 0.455$) [112].

structures seem to be related in comparison with Fig. 6, showing different stages in the formation of a bubble breakdown by core swelling. Fig. 19 is the latest stage of development, presented in Fig. 6f. Fig. 6c could be related to a double ring structure, where the compressed spiral (the outer ring) induces a cell of counter-rotating flow (the inner ring). This second ring is not visible as it is not dyed. So these different structures could exist at different stages of evolution or under different conditions. As the 'bubble' grows (the slope of Brücker's spiral

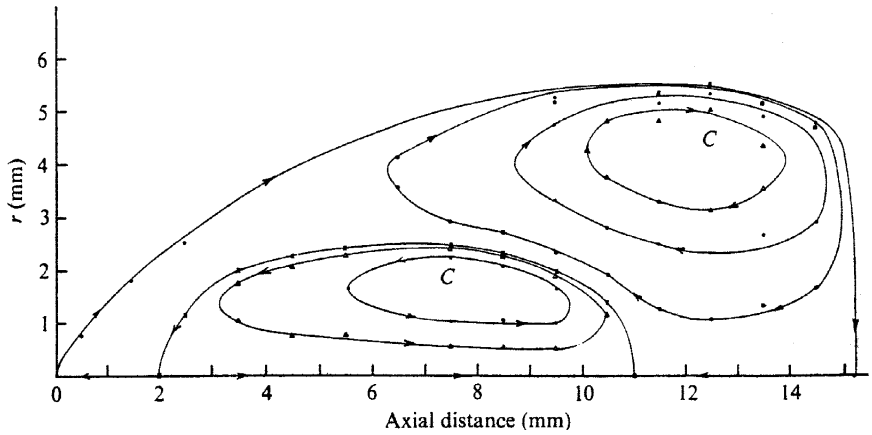


Fig. 18. Double ring structure [12].

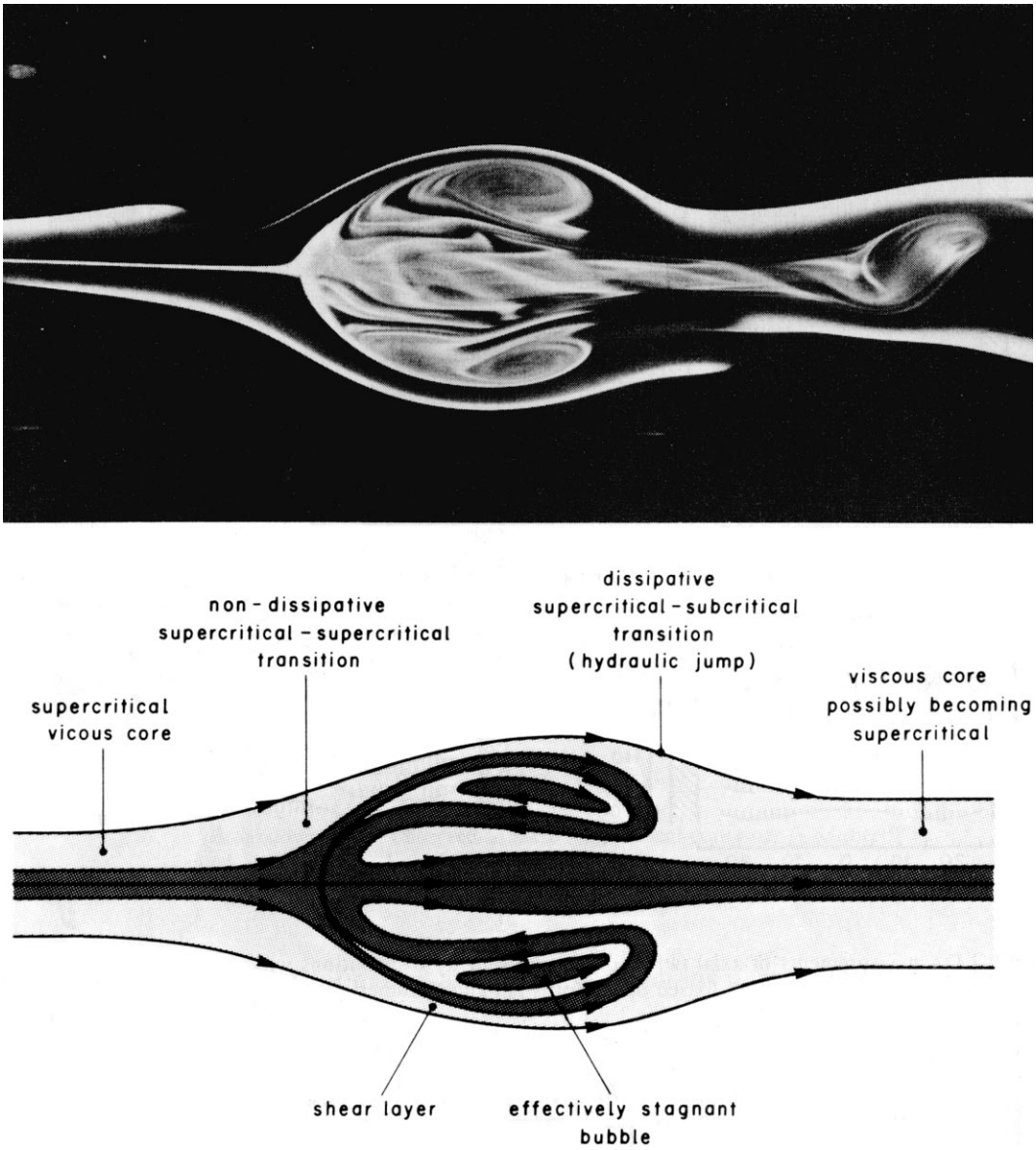


Fig. 19. Photograph and schematic representation of a bubble breakdown [16].

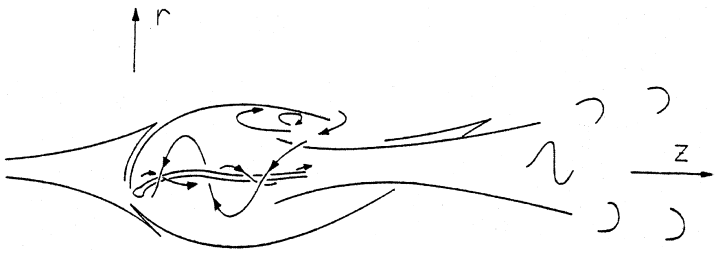


Fig. 20. Schematic diagram of the flow pattern in a bubble [58].

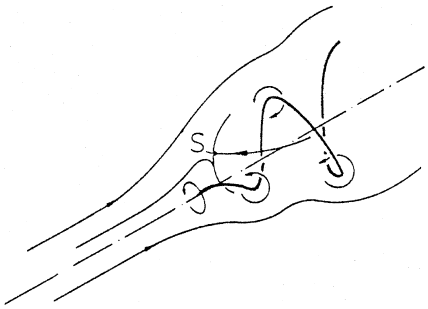


Fig. 21. Schematic view of the position of the stagnation point S in relation to the spiralling core, induced reversed flow and outer stream surfaces [39].

increases), the second inner ring, initially induced by the compressed spiral, would become more important and finally, would drain the recirculated flow towards the downstream region.

2.7. Overview of the experimental investigations

Different types of vortex breakdown have been observed for different combinations of Reynolds number Re_{tube} and circulation number (e.g. [10,17]). It has to be remembered that the dependence on Re_{tube} has to be regarded as a *vorticity* effect, not a *viscosity* effect.

All breakdown forms, including the bubble breakdown, are periodic and asymmetric in azimuth. The approach flow usually presents a jet-like axial velocity profile, whereas the flow downstream of the breakdown is wake-like and (more) turbulent with strong periodic oscillations. This is reminiscent of the wake of a solid body revolution.

At high Re_{tube}, the only forms observed are the bubble and the spiral. The bubble mode of disruption of the vortex core is characterised by a stagnation point on the swirl axis, followed by a bubble of recirculating fluid. The spiral breakdown is also characterised by a rapid deceleration of the flow causing stagnation and, in addition, by an abrupt kink followed by a spiral twisting of the flow. The stagnation point appears to be off-centred. The winding of the spiral has been observed to occur in the sense of rotation of the outer flow (e.g. [10,11,17]) or opposite to it [13,38,39]. No definite explanation has been provided so far to explain this discrepancy. The spiral rotates as a structure in the sense of the base flow. Another discrepancy between experiments concerns the core size downstream of breakdown: the vortex core either appears expanded and weaker, the expansion being significantly larger for the bubble form than for the spiral [15,37] or the core radius is reduced and the axial vorticity intensified [59].

In the progression of forms as the swirl is increased, spiral breakdown always occurs before bubble breakdown. The spiral form also appears to be located further downstream than that of the bubble form. Depending on the combination of Re_{tube} and circulation, the bubble may evolve from a spiral, or directly from an axisymmetric swelling of the vortex core. There are ranges of flow settings for which spiral and bubble can exist and transform spontaneously into each other.

From Refs. [10,11,29,46], it appears that the occurrence of the breakdown and its location when it does occur depends, among other things, on the magnitude of the flow and the swirl, the external pressure gradient, the degree of divergence of the flow, and the initial upstream conditions. It is also strongly dependent upon the confinements placed upon it.

The frequency and the amplitude of the unsteady wall pressure have been shown to increase rapidly with increasing swirl.

Quantitative measurements described the interior of the bubble breakdown as a two-celled structure with positive axial velocity near the centreline or as a single tilted ring-like structure with an overall flow reversal near the centreline. The bubble seems to exchange fluid continuously with the outer flow through its downstream end.

Ideas, which are discussed in more detail later, also diverge regarding the fact that bubble and spiral forms of breakdown are different aspects of a same phenomenon or are two distinct phenomena.

Table 1 summarises, chronologically, some of the numerous publications related to experimental studies of the vortex breakdown phenomenon. The table mainly refers to flows *in tubes* and contains essentially the references used in this review. It is far from being exhaustive. Each entry consists of the year of publication, the reference number in the thesis, the name of the author(s), the type of equipment, flow, and technique, and the principal feature(s) investigated. Similar tables will be presented later for the numerical simulations and the theoretical investigations.

3. Numerical computation of vortex breakdown

3.1. Introduction

The phenomenon of vortex breakdown is extremely complex and it is quite difficult to obtain experimentally enough details to fully understand it and comprehend the mechanism(s) involved. With the development of computers and numerical techniques, a computational approach seemed very attractive to obtain additional information concerning the structure of the breakdown, as well as identifying the various parameters affecting its occurrence and development. Getting additional insight into the physics itself of the phenomenon has been another ambition.

The modelling of highly swirling flows is a difficult task, both for physical and numerical reasons due to the high degree of coupling appearing between the momentum equations when the influence of the rotational terms is large. Another particular difficulty is the upstream influence in

Table 1
Chronological list of some experimental investigations on vortex breakdown

Year	Author(s)	Equipment/flow/technique	Principal features
1954	Vonnegut	Tangential inlet into a cylindrical cavity with downstream tube (vortex whistle) — air and water	Time-dependent periodic motion
1957	Peckham and Atkinson	Delta wing	First observation of breakdown of leading-edge vortices
1961	Lambourne and Bryer	Delta wings	Spiral structure — frequency $= f$(flow rate)
1962	Harvey	Straight tube — guide vanes	Only bubble breakdown — effect of core size
1963	Chanaud	Tangential inlets — vortex whistle — air and water	Periodic fluctuations
1964	Lowson	Delta wings	Position of breakdown $= f$(pressure recovery)
1965	Chanaud	Tangential inlets — vortex whistle and cyclone separator — air and water	Descriptive study of temporally periodic motion
1967	Hall	Theoretical approach	Axial pressure gradient $= (P_{\mathrm{grad}})_{\mathrm{externally\ imposed}} + (P_{\mathrm{grad}})_{\mathrm{swirl}}$
1967	Lambourne and Bryer	Delta wing	Two forms of breakdown observed: axisymmetric and spiral — spiral's winding *opposite* to the surrounding fluid sense of rotation — adverse pressure gradient on an inviscid model vortex leads to reverse flow
1970	Cassidy and Falvey	Straight tube — guide vanes — smoke at low *Re* and velocity/*P* measurements at higher Re^{a}	Unsteady helical vortex — correlations in dimensionless form
1971a,b	Sarpkaya	Diverging cylindrical tube — guide vanes	Three types of breakdown: double helix, spiral and bubble — breakdown location $= f(Re^{a}, \Omega^{b})$ — both bubble and spiral can exist in some ranges — rear end of bubble open for fluid exchange — spiral's winding *with* the surrounding fluid — multiple breakdowns
1972	Hall		Review
1974	Sarpkaya	Diverging cylindrical tube — guide vanes	Effect of adverse pressure gradient and degree of divergence — breakdown location $= f(Re^{a}, \Omega^{b}, \gamma^{c})$
1977	Faler and Leibovich	Flared tube — guide vanes	Seven types of vortex breakdown — breakdown location $= f(Re^{a}, \Omega^{b})$ — different forms can exist in some ranges — fluid exchange
1978	Faler and Leibovich	Flared tube — guide vanes — LDA	Two-celled structure in the bubble type of breakdown with axial velocity oriented downstream in the interior region — highly periodic velocity fluctuations in the downstream half of the bubble and in the wake
1978	Leibovich		Review
1979	Garg and Leibovich	Flared tube — guide vanes — LDA — power spectra	Jet-like approach flow and wake-like downstream flow — expansion of the vortex core — coherent low frequency oscillations more energetic for the bubble type
1980	Escudier, Bornstein and Zehnder	Cylindrical tube — slit inlet — range of exit diameters — LDA	Change in the vortex structure as the exit diameter is reduced
1982	Escudier and Zehnder	Cylindrical and flared tubes — slit inlet	Criterion for the occurrence of breakdown $= f(Re^{a}, \Omega^{b}, R^{d})$ — 'prebreakdown' phenomenon — sense of winding of the spiral with the outer fluid (prebreakdown) or opposite (breakdown)
1984	Escudier	Cylindrical container with rotating endwall	Only bubble breakdown
1984	Leibovich		Review

Table 1 (*continued*)

Year	Author(s)	Equipment/flow/technique	Principal features
1985	Escudier and Keller	Straight tube with different exit contractions — guide vanes — water — LDA	Upstream influence of the exit contraction is the flow is subcritical between breakdown and exit
1985	Krause	Theoretical work	Breakdown location = f(core growth, w_{max},[e] u_∞[f])
1985	Staufenbiel and Helming	Delta wing — water tunnel	Breakdown reversed to normal if negative P_{grad} applied
1985	Uchida et al.	Pipe — guide vanes — LDV	Sketch of the flow pattern inside the bubble
1987	Hallett and Toews	Sudden expansion, tangential inlets, high *Re*	$S_{cr} = f$(inlet velocity profiles, expansion ratio) — low u near the axis or small solid body vortex core decrease S_{cr} required for reverse flow — increase in expansion ratio reduces S_{cr} until a ratio ~1.5
1987	Nakamura and Uchida	LDA	Velocity profiles *upstream* of a bubble and a spiral breakdown
1987	Stuart		Review
1988	Escudier	Combustor geometry from Ref. [55]	Effect of pressure gradient — influence of exit contraction
1988	Payne et al.	Delta wings	Periodic transition between bubble and spiral forms
1989	Farokhi, Taghavi and Rice	Manifold rings and elbow nozzles	Effect of initial swirl distribution for same S,[g] Ma[h] and Re[a] — S insufficient to describe the character of swirling flows
1991	Chao et al.	Swirl combustor — guide vanes — LDA — hot wire anemometry — microphone transducer	Downstream contraction favours breakdown — 2 dominant natural frequencies (PVC and downstream azimuthal instability) — the downstream azimuthal instability promotes the breakdown recirculation
1992	Brucker and Althaus	Cylindrical tube — guide vanes — PTV + volume scanning — water	Bubble structure — single vortex ring tilted against the centreline and gyrating — vortex axis remains at the centreline
1993	Brucker	Cylindrical tube — guide vanes — PTV + volume scanning — water	Spiral structure — vortex axis off-centred — difference between modes explained by changes of spiral's slope, winding and diameter
1995	Althaus et al.		Review
1996	Gursul		Sense of winding of the spiral = f(axial velocity profile)

[a] *Re* based on tube diameter.
[b] $\Omega = \Gamma/u.D$ = circulation number.
[c] γ = degree of divergence.
[d] $R = Q/\Gamma.L$ (ratio v/w in the inflow region).
[e] w_{max} = maximum azimuthal velocity.
[f] u_∞ = freestream axial velocity.
[g] S = swirl number.
[h] Ma = Mach number.

subcritical flows, underlined by Escudier [16], which requires an accurate definition of the boundary conditions at the exit. While efforts have been, and still are, made to improve the turbulence models, this last question has often been neglected.

The assumptions, equations, procedure, etc. used are not detailed in the text, but are summarised in the chronological Table 2. The investigations presented here only deal with incompressible flows.

Again, this review mainly concerns vortex breakdown *in tubes*. However, some interesting or striking results regarding modelling of vortex breakdown above delta wings and in closed containers with a rotating endwall will be mentioned.

3.2. Quasi-cylindrical approximation

Upstream of the breakdown, the axial gradients are small compared with the radial gradients, and the stream surfaces are approximately cylindrical if the vortex is assumed to be slender, i.e. if the ratio of the core radius to the breakdown length is much smaller than unity. Quasi-cylindrical equations, discussed in Ref. [43], can then be used to calculate the development of the flow: if the velocity distributions are given at some initial upstream location, the solution can be computed by proceeding step by step axially downstream, similarly the calculation of two-dimensional boundary layers. The idea is that the development of large axial gradients associated with vortex breakdown violates the assumptions of the quasi-cylindrical approximation (QCA), in the same way the approach to separation violates the boundary-layer approximation, so that the failure of the QCA corresponds to vortex breakdown. As Reynolds numbers in most experiments are sufficiently large to insure that viscous forces are negligible, viscosity is unlikely to cause the approximation to fail.

Different versions have been proposed, applying different assumptions, mainly on velocity profiles of the approaching flow: [42,60–62]. Usually, the flow has been assumed to be steady, laminar, incompressible, and axisymmetric.

The following velocity profiles have been commonly adopted [16]:

- Rankine vortex

  $u = \text{constant}$

  and

  $w = \Gamma_\infty r/2\pi\delta^2 \;\text{ for } r < \delta$

  $w = \Gamma_\infty/2\pi r \;\text{ for } r > \delta$

  where δ is the core radius and ∞ is a constant circulation
- Hall vortex

  $u = \Delta v(G - 1/2 \ln z)$

  $v = -1/2\Delta v\,(z)^{-1/2}$

  $w = \Delta v(G + 1/2 - 1/2 \ln z)^{1/2}$

  where Δv determines the magnitudes of the velocity components and G is a shape factor for the profile $z = (r/\delta)^2$
- Q vortex

  $u = \Delta u\, \mathrm{e}^{-z}$

  $w = q|\Delta u|\delta/r(1 - \mathrm{e}^{-z})$

  where Δu is a velocity scale (examples of evaluation in Fig. 22)

  $q = \Gamma_\infty/2\pi\delta\Delta u = 1.57\, w_{\max}/\Delta u$

  where $w_{\max}$ is the maximum swirl velocity
- Exponential vortex

  $u = \text{constant}$

  $w = \Gamma_\infty/2\pi r(1 - \mathrm{e}^{-z})$

Hall [42], using the QCA, provided an analogy between the initiation of vortex breakdown with that of the development of a separated region for a boundary layer subjected to an adverse pressure gradient. Hall backed this analogy by showing that the occurrence and position of the failure of the approximation depends on the level of swirl and degree of divergence of the tube (or positive pressure gradient) in just the same way as vortex breakdown does.

Krause [44] also demonstrated the crucial influence of the pressure field on the breakdown length of slender vortices. He also got some insight into the process of vortex breakdown as he compared computational results for inviscid and viscous flows. With inviscid flows, an externally imposed adverse pressure gradient was necessary for initiating breakdown, whereas the phenomenon could be initiated without any for viscous flows; however, if an external pressure gradient was imposed in this latter case, the breakdown length was largely reduced. From these numerical experiments, it appears that the pressure field indeed dominates the breakdown mechanism of slender vortices, but the viscous forces also play a role in initiating the widening of the core.

While *this numerical* method seems to be able to predict the occurrence and position of breakdown, it is, however, limited as it cannot account for upstream influences and cannot describe the internal structure of the breakdown nor the flow downstream of it. A more complete simulation of vortex breakdown requires the full Navier–Stokes equations.

3.3. Solutions to the Navier–Stokes equations

Although technological applications (aeronautic, combustion, etc.) invariably involve turbulent flows and the phenomenon is essentially asymmetric, even in the

Table 2
Chronological list of some contributions to numerical simulation of vortex breakdown

Year	Author(s)	Flow/assumptions	Equations	Main conclusions
1972	Hall		QCA	Review and criticism
1973	Kopecky and Torrance	Inviscid, axisymmetric	NS equations — circulation/vorticity formulation—cylindrical coordinates	Retardation, stagnation, bubble-like structure
1976	Grabowski and Berger	Unconfined viscous vortex, laminar, steady, axisymmetric	NS equations — velocity/pressure formulation—artificial compressibility—cylindrical coordinates	Retardation, stagnation, bubble-like structure—diffusion and convection of vorticity away from the core
1985	Krause	Slender vortices—viscous/inviscid	QCA	Viscous forces initiate the widening of the core—pressure field decisive for the breakdown length
1985	Shi	Isolated axisymmetric concentrated vortex, unsteady	NS equations — vorticity/stream function formulation—cylindrical coordinates	Double ring structure—periodic flow behaviour
1987	Hafez, Ahmad, Kuruvila and Salas	Inviscid/laminar, axisymmetric, steady/unsteady	NS equations — vorticity/stream function formulation—cylindrical coordinates	Multiple breakdowns—unstable solution at high Re_{core}[a]
1987	Krause and Menne	Incompressible/compressible, axisymmetric, unsteady	NS equations — vorticity/stream function formulation—cylindrical coordinates	Influence of inflow boundary cond.—double ring structure—periodic flow behaviour
1987	Spall and Gatski	Laminar, 3D, time-dependent	NS equations — velocity/vorticity formulation—Cartesian coordinates	Bubble breakdown—single ring—steady—vorticity transfer (from axial to other comp.) and intensification—*Ro* criterion
1989, 1993	Breuer and Hanel	Laminar, 3D, time-dependent	NS equations — artificial compressibility and dual time—stepping—Cartesian coordinates	Bubble: 2-rings structure — reaches the inflow section — spiral breakdown: winding opposite to basic flow — type $=f(P_{grad})$
1989	Salas and Kuruvila	Laminar, axisymmetric flow-steady and unsteady calculations	NS equations — vorticity/stream function formulation — cylindrical coordinates	Multiple breakdowns — Hopf bifurcation suggested
1990a, b	Krause	3D, time-dependent	NS equations — artificial compressibility and dual time-stepping — Cartesian coordinates	Bubble and spiral breakdown
1990	Lopez	Laminar, axisymmetric, unsteady [closed container with rotating endwall]	NS equations — stream function/vorticity formulation — cylindrical coordinates	Multiple recirculation zones — very good agreement with experiment — negative azimuthal vorticity generated
1990	Spall and Gatski	Laminar, 3D, time-dependent	NS equations — velocity/vorticity formulation — Cartesian coordinates	Type of breakdown $=f(P_{grad})$ — vorticity transfer
1990	Spall, Gatski and Ash	Laminar, 3D, time-dependent	NS equations — velocity/vorticity formulation — Cartesian coordinates	Bubble externally axisymmetric and steady — interior asymmetric and periodic with a 2-rings structure — vorticity transfer and intensification
1991	Gatski and Spall	Laminar, 3D, time-dependent	NS equations — velocity/vorticity formulation — Cartesian coordinates	Spiral and double helix when decreasing pressure gradient
1991	Spall and Gatski	Laminar, 3D, time-dependent	NS equations — velocity/vorticity formulation — Cartesian coordinates	Four types of breakdown (weak helical, double helix, spiral and bubble) — impact of freestream axial velocity on position and type
1992	Aksel and Kaya	Laminar, axisymmetric	Finite element	General properties of axisymmetric vortex breakdown
1992	Spall, Gatski and Kandare	Laminar, 3D, time-dependent	NS equations — velocity/vorticity formulation — Cartesian coordinates	Spiral, double and simple helix when decreasing P_{grad} — irrotational region deflected at breakdown location except for bubble
1995	Spall and Gatski	turbulent, 3D, unsteady	Reynolds averaged NS equations-velocity/vorticity formulation — Cartesian coordinates	Single bubble-type breakdown — no negative vorticity, lower rate of decay

[a] Re_{core}: Reynolds number based on the core diameter.

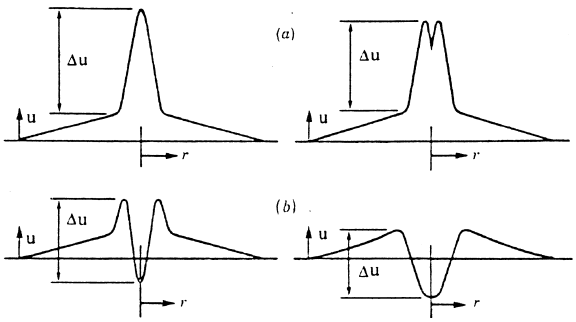


Fig. 22. Definitions of Δu for stability calculations of the measured velocity profiles: (a) jet-like; (b) wake-like [91].

bubble form, most numerical computations of vortex breakdown were initially restricted to incompressible, laminar, steady and axisymmetric flows due to the complexity of the problem and computational constraints. Time-dependence, three-dimensionality and turbulence were dealt with later.

The first numerical solutions to the steady axisymmetric equations for vortex breakdown were presented by Kopecky and Torrance [63] and Grabowski and Berger [64]. These were broadly similar, as stated by Leibovich [7]: for a given flow rate, deceleration on the axis and local swelling of the stream lines were obtained at low swirl. At higher values, a stagnation point appeared on the axis followed by a small region of closed streamlines resembling a bubble. Further increasing the swirl caused the bubble to grow in size. Grabowski and Berger obtained, for some cases, a second region of reversed flow, which was never observed in Kopecky and Torrance's results. The coarser grid used by Kopecky and Torrance could be at the origin of this difference. The description seems to agree with some experimental observations. However, a lot of discrepancies exist [7]:

- The mild bulges obtained at low swirl were never observed experimentally, except as transients. Bubbles smaller than the approaching core radius was also never reported. At low swirl, spiral structures were observed instead.
- Only very slight core expansions were computed, not in relation with the experimental observations.
- The double ring structure observed experimentally in the interior of the bubble by Faler and Leibovich [17] was not reproduced.

Shi [65] presented time-dependent, axisymmetric results which clearly revealed a two-celled structure within the bubble breakdown, together with a periodic flow behaviour. He did not report multiple breakdown. Krause and Menne [66] also obtained this type of solution when the axial gradient of the radial velocity was set equal to zero in the inflow section ($dv/dx = 0$), instead of assuming an advancing parallel flow ($v = 0$), like Refs. [63,64]. This type of boundary condition might have been required because of the lack of proper approaching flow. One bubble was first generated, containing one vortex ring and, some time later, a second, counter-rotating vortex ring appeared. A structure resembling that observed experimentally was obtained for a certain time. However, the structure moved upstream in time and finally reached the inflow cross-section. These initial results seem to show that vortex breakdown could be simulated numerically, provided the vortex structures do not travel upstream to the inflow cross section. However, Spall and Gatski [67] criticised Krause and Menne's results as *the bubble had lifted off the axis*.

Multiple breakdowns have been observed experimentally by Harvey [8] and Sarpkaya [11] and have been obtained numerically by Hafez et al. [68] and Salas and Kuruvila [69]. In their calculations, as the Reynolds number Re_{core} (based on the vortex core radius) was increased from a low value, more and more breakdown bubbles developed. The azimuthal vorticity intensified where breakdown was to occur, signalling the appearance of the bubble. The flow structure was also very sensitive to small variations of the swirl parameter. These behaviours are illustrated in Figs. 23 and 24. These authors also considered unsteady solutions, starting from steady solutions randomly perturbed. For low Re_{core} (only one breakdown), steady-state solutions of the unsteady equations were identical to those of the steady equations. For higher Re_{core} (two breakdowns), time-dependent calculations did not reach a steady-state but rather oscillated around it. Salas and Kuruvila [69] then suggested the existence of a Hopf bifurcation (transition from a steady to an unsteady, periodic flow).

Similar results were obtained by Aksel and Kaya [70] using a finite element method: for high values of Re_{tube}, as observed in the experimental works ($1000 < Re_{tube} < 6000$), the numerical solutions did not converge. For these high Re_{tube} the convergence of numerical solutions could only be obtained for small circulation numbers. They suggested that this *may be due to the fact that the convective terms in the governing equations start to dominate the viscous diffusion terms for larger Re so that the nonlinear character will be more dominant*. For large circulation numbers, a second bubble was not computed, although the presence of mild bulges downstream of the bubble breakdown suggested it. This was attributed to the relatively coarse mesh.

Multiple recirculation zones were also reproduced numerically in cylindrical containers with a rotating endwall by Lopez [71] and accorded faithfully with the experimental visualisations of Escudier [14] as shown in Fig. 25. The attraction of this approach to vortex breakdown is that the flow is defined by only four variables (height and radius of the cylinder, angular velocity and kinematic viscosity) each of which can be very accurately determined experimentally. Numerically, the problem is also easier as the flow is confined in a fixed volume with very well-defined boundary

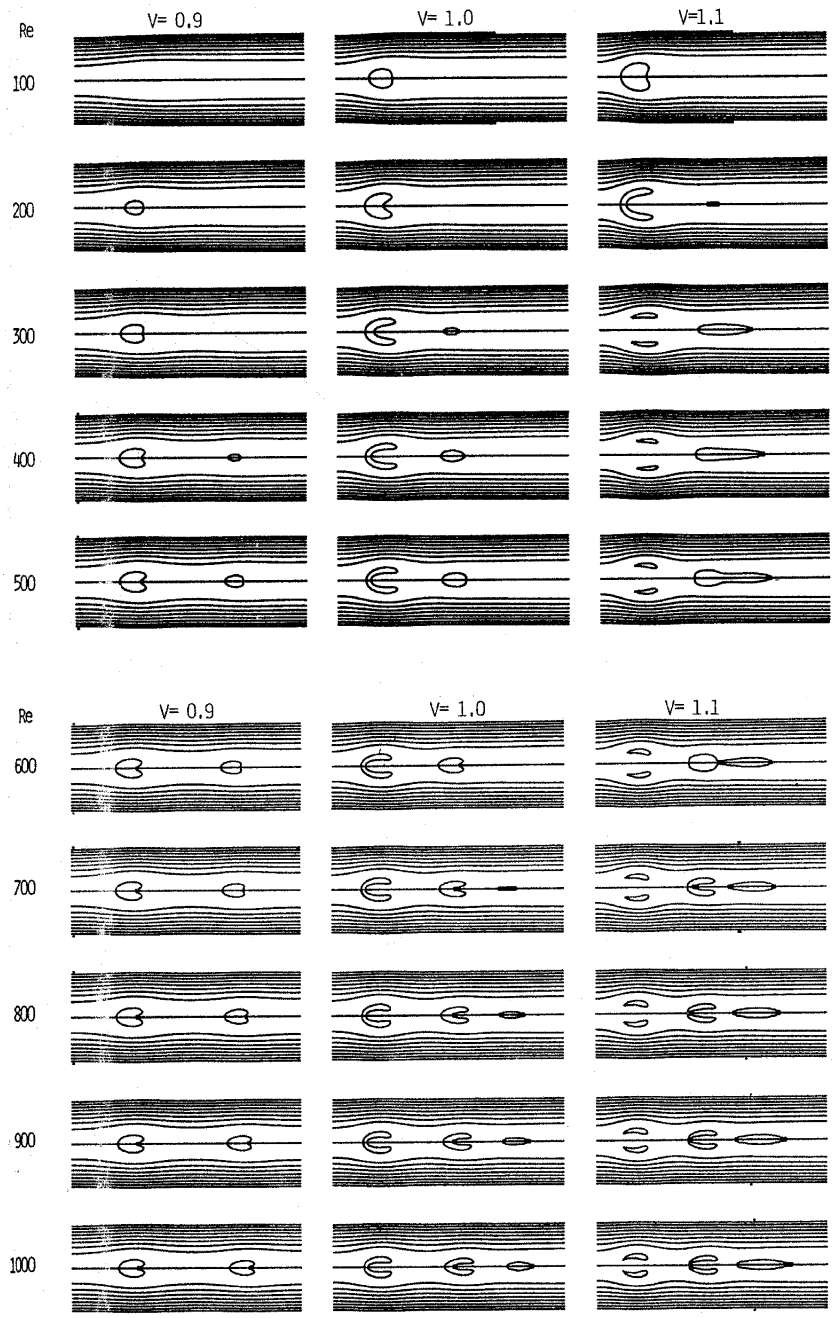


Fig. 23. Computed streamlines for different Reynolds numbers and swirl levels.

conditions. Lopez concluded that the appearance of the recirculation zones was associated with the development of a negative azimuthal component of vorticity, i.e. which induces an axial velocity component opposite to the base flow.

A number of these simulations were made under the assumption of axisymmetric flow. However, as pointed out by Leibovich [7], the spiral form of breakdown was excluded by the hypothesis of axisymmetry and, even for the bubble form of breakdown, important structural features were not accessible or were incorrectly represented. Spall and Gatski [67] were the first, to the author's knowledge, to present a three-dimensional, time-dependent simulation of vortex breakdown. This numerical study was performed

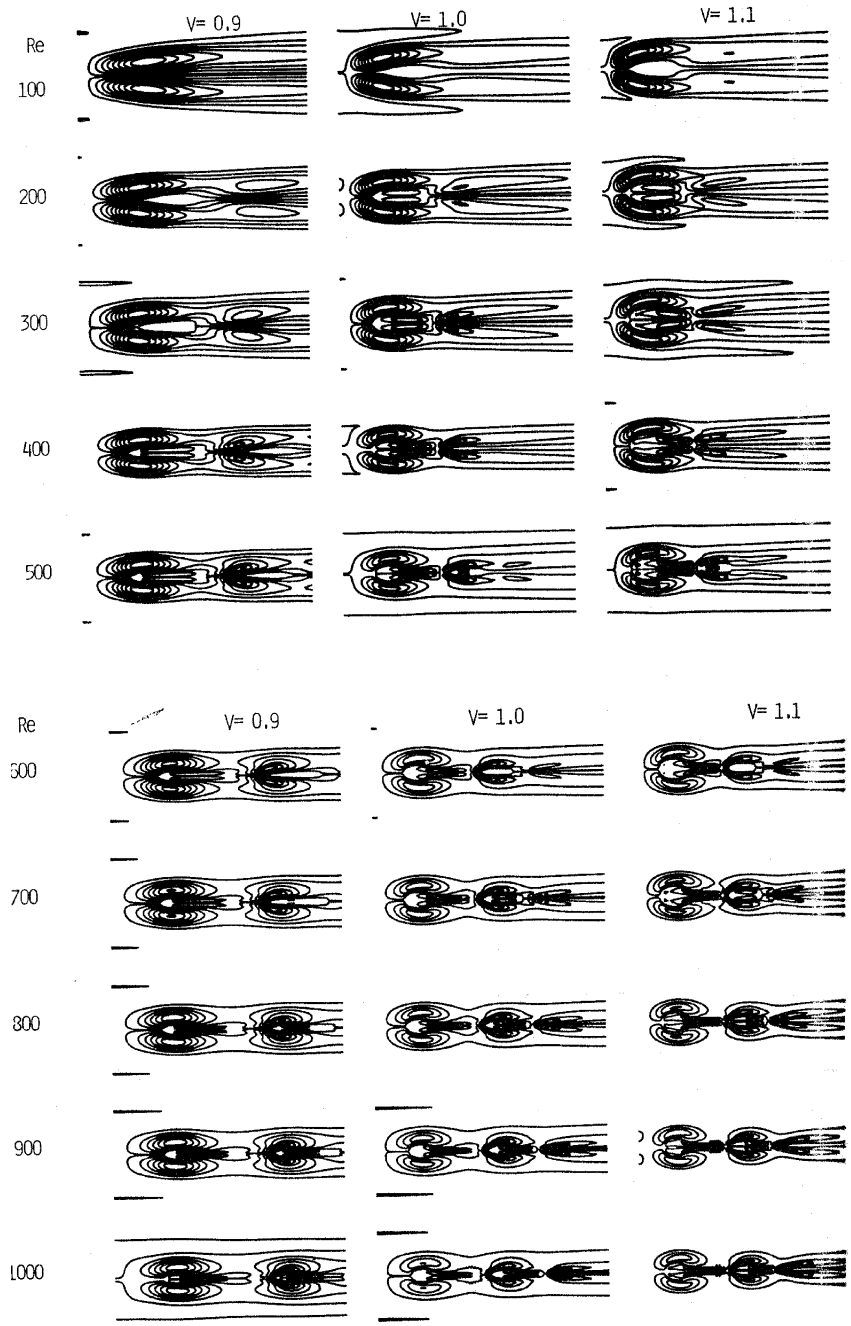


Fig. 24. Computed azimuthal vorticity for different.

taking into account a criterion developed by Spall et al. [4] on the basis of the theoretical works of Squire [72] and Benjamin [51] and will be discussed later. To summarise, it can be said that the criterion is based on a Rossby number $Ro = u/r_{\text{core}}\Omega$ (Ω being the vortex core rotation rate). This nondimensional parameter represents the ratio of a characteristic axial velocity to a characteristic swirl velocity and is a measure of the relative importance of the inertial and Coriolis forces on the fluid motion. The criterion defines, for a given Re_{core}, the critical Ro below which vortex breakdown will occur. They used a value slightly under the critical one for their simulation. The study concentrated on the breakdown structure, but also analysed the effect of vortex breakdown on the vorticity distribution. A local

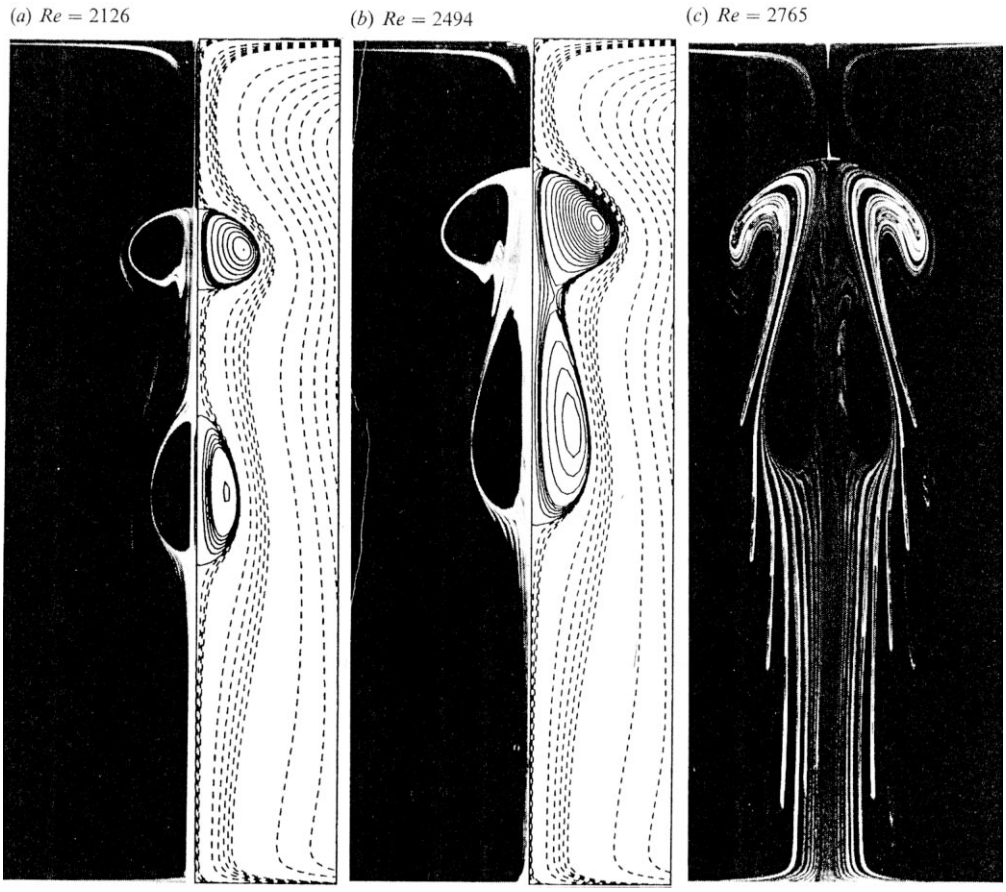


Fig. 25. (a) and (b) computed contours of the stream function (right halves) compared with visualisations of steady dye-lines (left halves); (c) snap-shot of the unsteady dye-lines in an oscillatory flow (from Ref. [71]).

intensification of enstrophy (i.e. the total 'amount' of vorticity) was noticed inside the simulated bubble breakdown, *due to the stretching of the vortex lines*. The axial component itself, the only one present initially, was small within the breakdown region while radial and azimuthal components appeared. This transfer of vorticity was explained by the tilting of the vortex lines. The flow downstream of the breakdown was nearly quasi-cylindrical, with a small axial velocity deficit and the core was expanded. A local intensification of the axial component of vorticity just aft of the breakdown region was noted. The computed bubble breakdown appeared quasi steady and symmetric and consisted of a single recirculation zone, which did not agree with Faler and Leibovich's experiments [7] where a double ring structure and periodic asymmetric oscillations were observed. The authors attributed these discrepancies to the artificial boundaries (uniform free stream instead of slightly diverging duct, vortex breakdown occurring near the inflow boundary). It was also suggested that the *Ro* number of the base flow near the inflow should be above the critical value, and then decreased as the flow progresses downstream, to simulate a region of approaching flow before breakdown.

Applying these recommendations, Spall et al. [73] predicted a bubble, axisymmetric in appearance, but with a clearly asymmetric internal structure (Fig. 26).

The qualitative agreement with the experiments of Faler and Leibovich [12] was judged excellent (compare Fig. 26 with Fig. 17). The inflow vorticity was specified symmetric. Within the breakdown bubble, the level of axial vorticity was reduced due to the vortex tilting causing a redistribution of the axial vorticity to the other components. This transfer took place as the first stagnation point was approached. For strong deceleration of the flow, regions of negative vorticity developed. The azimuthal vorticity reached maximum levels near the axial location at which the breakdown was initiated, but off the centreline. Downstream of the breakdown, the vorticity distribution was again nearly symmetric, although more intense than at the inflow. It was concluded that, through vortex stretching, the breakdown bubble intensified axial vorticity (Fig. 27). The swirl velocity distribution downstream of the breakdown also showed

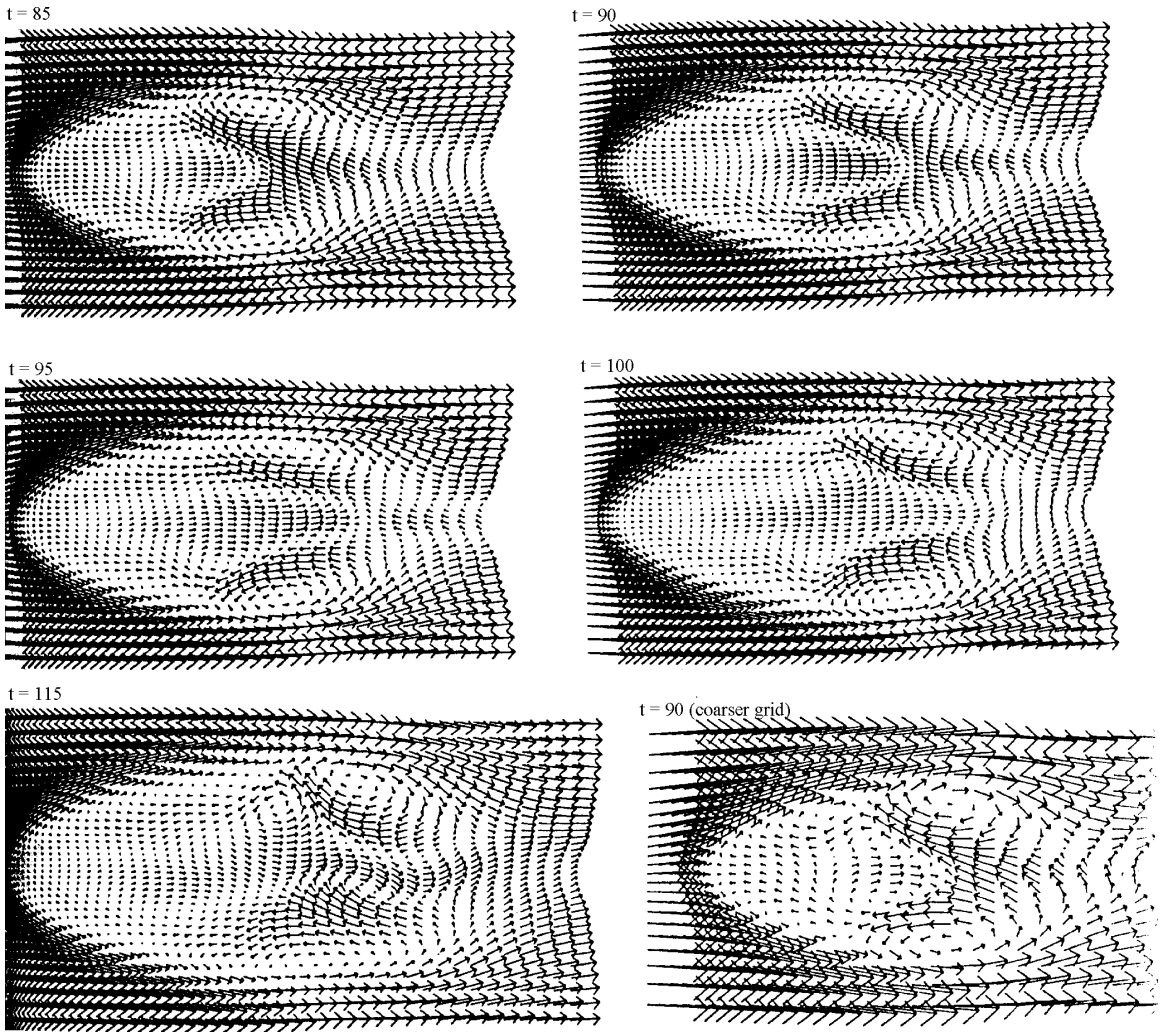


Fig. 26. Velocity vectors over the interior of the breakdown region at different times (from Ref. [73]).

magnitude intensification based on the local characteristic rotation rate (i.e. the slope of the swirl velocity profile at the centreline) and reduction of characteristic size. This last result agrees well with Brücker and Althaus' measurements [59], but is not coherent with the core expansion observed experimentally by Faler and Leibovich [12].

This discrepancy has been commented on by Spall et al. [73]: in both studies, the local characteristic rotation rate increased relative to the characteristic rates upstream of the breakdown. However, the swirl velocity profile in Faler and Leibovich's work presented two local maxima, one giving a smaller core radius than upstream and a second, higher, giving a larger core radius (Fig. 28), whereas Spall et al.'s velocity distribution clearly shows only one maximum, although a second one was suggested (Fig. 29).

They attributed these differences possibly to different lateral boundary conditions (tube walls for Faler and Leibovich and free field for Spall et al.) or enhanced mixing in the tube (existence of a transitional flow condition) not present in the laminar flow of Spall et al. Another point brought up by the authors concerned the low Re_{core} used (200) which corresponds to a regime in which viscosity plays an important role contrary to the physics of most applications. However, the grid resolution was a limitation and the effects of turbulence could not be examined at that stage.

Breuer and Hänel [74,75] and Krause [76,77] also performed three-dimensional, time-dependent simulations of vortex breakdown. Their results showed a swelling, in time, of the vortex core, combined with the development of a free stagnation point. A bubble evolved, grew, moved upstream as observed experimentally by Escudier [16]. The internal structure, observed by Breuer and Hänel through the vorticity distribution, consisted initially of one

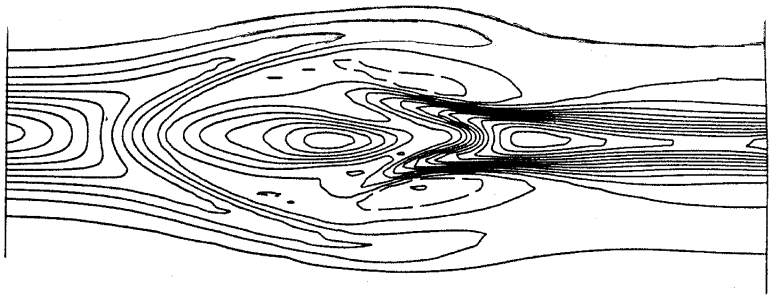


Fig. 27. Contours of axial vorticity [73].

ring (Fig. 30). In time, a second counter-rotating ring formed inside the first one.

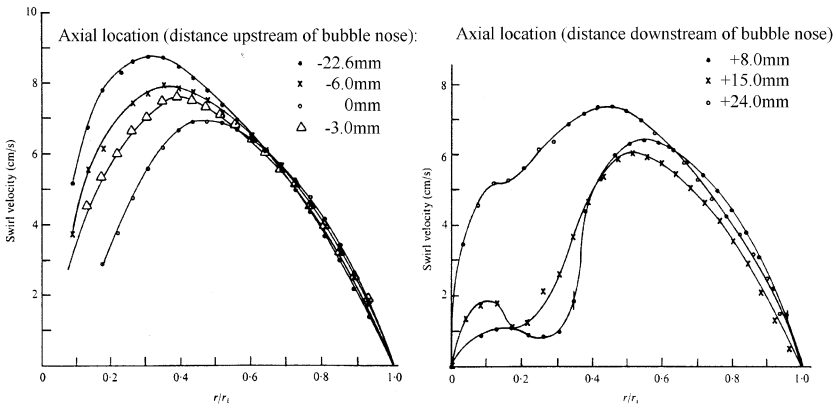
While this structure agrees qualitatively with Faler and Leibovich's experiments [7], the authors commented that *these results must be interpreted with caution since the bubble had already reached the inflow section*. After stabilising the breakdown location by reduction of the imposed pressure gradient on the outer flow and by adapting the lateral boundary conditions with time (iterative procedure presented in Ref. [76], the different authors observed, at larger times, a highly asymmetric solution (Fig. 31). The structure represented by the streaklines looked like a spiral breakdown (Fig. 32). The sense of winding of the spiral was opposite to that of the basic flow, whereas the spiral itself precessed with the flow. Breuer and Hänel concluded that *the mode of breakdown seems to depend on the imposed pressure gradient*.

By adjusting the level of flow deceleration assigned at the lateral boundary, Spall and Gatski [78,79] and Spall et al [80] were also able to study the structure of different types of breakdown: simple helix, double helix, spiral and bubble. Consistent with the experimental observations, the numerical calculations highlighted an upstream movement of the different structures as adverse pressure gradient was increased. However, they did not show any tendency for an abrupt change from one type to another. Rather, the transformation appeared, as in Ref. [74], to be a gradual process resulting from an increase in the adverse pressure gradient. This can be observed in the plots of planar projected velocity vectors presented in Fig. 33 for different adverse pressure gradients. The sequence reveals a clear transition ranging from a mild disturbance of the vortex core off the axis (Fig. 33a) similar to type 6 in Faler and Leibovich's [17] classification) to the bubble-type breakdown (Fig. 33d) similar to the type 0 breakdown described in Ref. [17]). Fig. 33c corresponds to a spiral breakdown (type 2). Athough not apparent from the velocity vector plots, Fig. 33b is characteristic of the double-helix breakdown.

The plots of constant normalised helicity density, presented in Fig. 34, also show this progression. The helicity density, defined as $u.\omega$, is a measure of the degree of alignment between velocity and vorticity vectors.

The spiral structure obtained (Fig. 33c) agrees well with the experimental results presented by Brücker [39] (Fig. 35).

An off-centre stagnation point has also been displayed. As also observed experimentally, the spiral breakdown became turbulent after a few turns of the spiral. In the bubble case, the vortex core was revealed weaker downstream of the breakdown, which is consistent with Faler and Leibovich's [12] conclusion and contrasts with their first calculations



(a) upstream (b) downstream

Fig. 28. Radial swirl vel. Profiles; $Re = 2560$, $\Omega = 1.777$ [12].

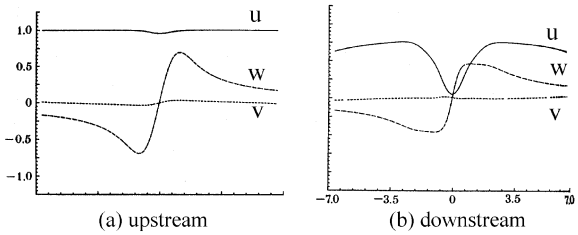


Fig. 29. Radial velocity profiles (Spall et al. [73]).

where vorticity amplification was reported [73]. For all the types of breakdown studied by Spall and Gatski [79], breakdown appeared to be *signalled by the formation of an irrotational region of fluid along the vortex axis*, i.e. as progressing downstream, the level of vorticity decreases along the vortex centreline to nearly zero at the location where the first signs of asymmetries in the flow appear. This region of irrotational fluid is immediately deflected off the axis at the breakdown point, except for the bubble type of breakdown. In this case, the region rather expands in an annular shape as proceeding through the forward portion of the bubble. This is suggested in Fig. 36 which shows the contour plots of constant axial vorticity corresponding to the same conditions as the helicity density plots presented earlier (Fig. 34). As the adverse pressure gradient is increased (from (a) to (d)), zones of negative axial vorticity appear which are initially deflected ((b) and (c)) and seem to evolve towards an annular structure as the flow field tends to resemble a bubble breakdown.

Turbulent bubble vortex breakdown was recently

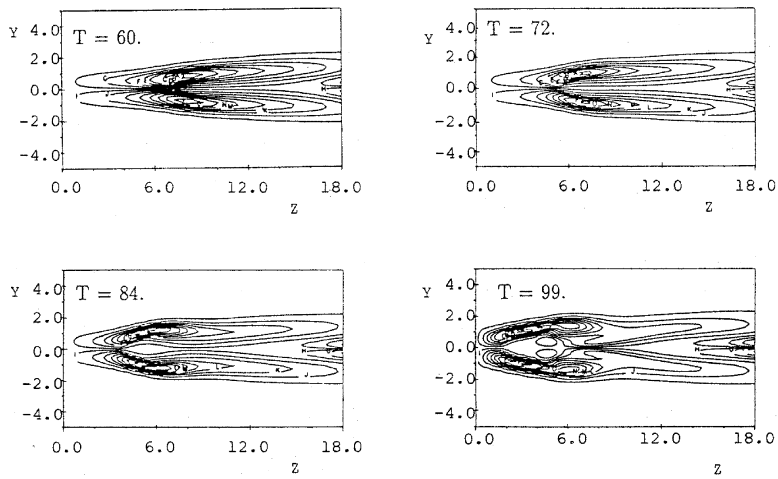


Fig. 30. Azimuthal vorticity contours at different times showing the formation of a second counter rotating vortex ring inside the first one [74].

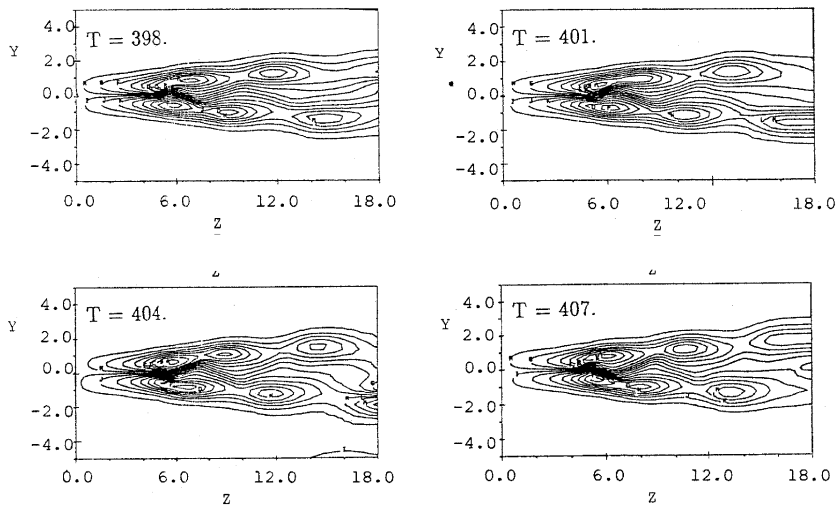


Fig. 31. Azimuthal vorticity contours at larger times showing asymmetries [74].

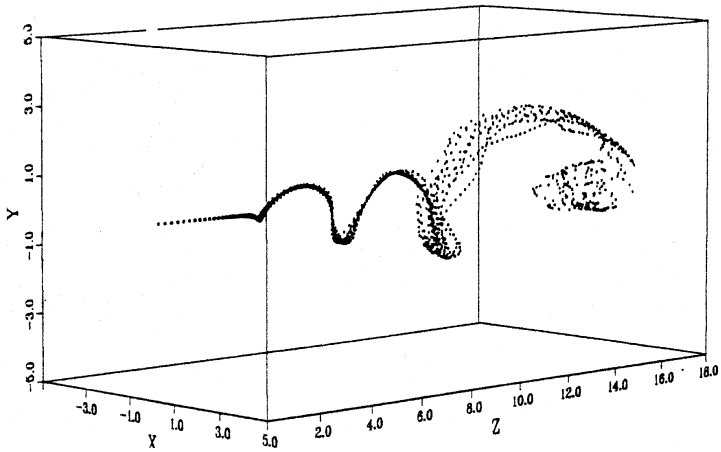


Fig. 32. Streaklines of an asymmetric solution [75].

examined by Spall and Gatski [81]. Two different turbulence models were employed: the standard k–ϵ and algebraic stress model (ASM). With the k–ϵ model, a higher level of lateral deceleration was required to initiate breakdown and the bubble produced presented a significantly smaller L/D ratio. The authors commented that these differences may be explained, respectively, by the excessive levels of turbulent diffusion in linear two-equation turbulence models and the tendency to underpredict the length of recirculation zones, in standard backward-facing step calculations and for swirling, recirculating flows in general [82]. In Fig. 37, the differences in the axial vorticity distribution between turbulent breakdown (computed using the ASM) and laminar breakdown are contrasted. Although the inflow distribution is identical in each case, the decay of axial vorticity, as the flow evolves in the streamwise direction, takes place more rapidly for the laminar case. The vorticity distribution is also simpler and regions of negative vorticity were not computed. Downstream of the vortex breakdown, the levels of vorticity in the vortex core are approximately three times lower for turbulent breakdown, which suggests that subsequent bubbles are much less likely to occur. This difference might also explain the discrepancy between the experimental observations of Brücker and Althaus [59] and Faler and Leibovich [12] regarding the axial vorticity intensity downstream of breakdown: at low Reynolds number ($Re_{\text{tube}} = 840$), Brücker and Althaus highlighted a smaller core with intensified vorticity, whereas at larger Reynolds number ($Re_{\text{tube}} = 2560$), Faler and Leibovich observed an expanded, but weaker core. A comparison with previous work by Spall and Gatski [73], however, is difficult as they used a Reynolds number based on the core diameter.

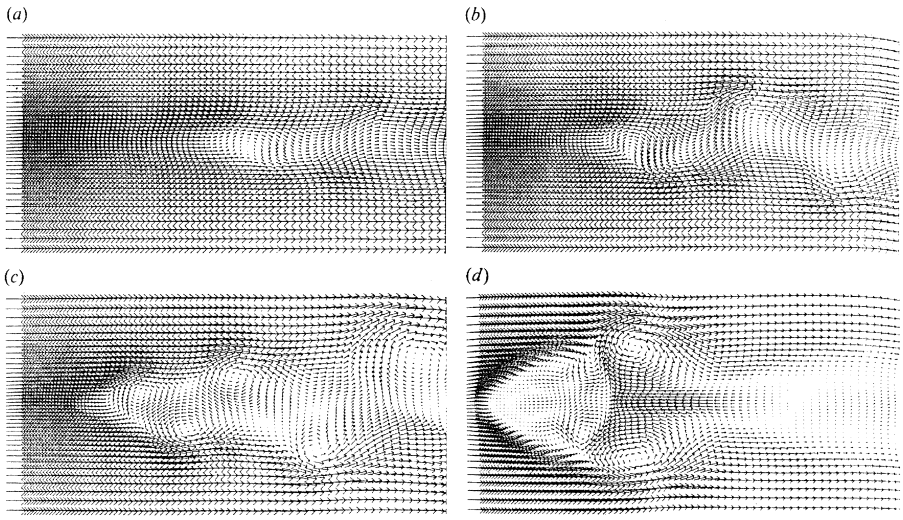


Fig. 33. Plots of planar projected velocity vectors on x–y plane.

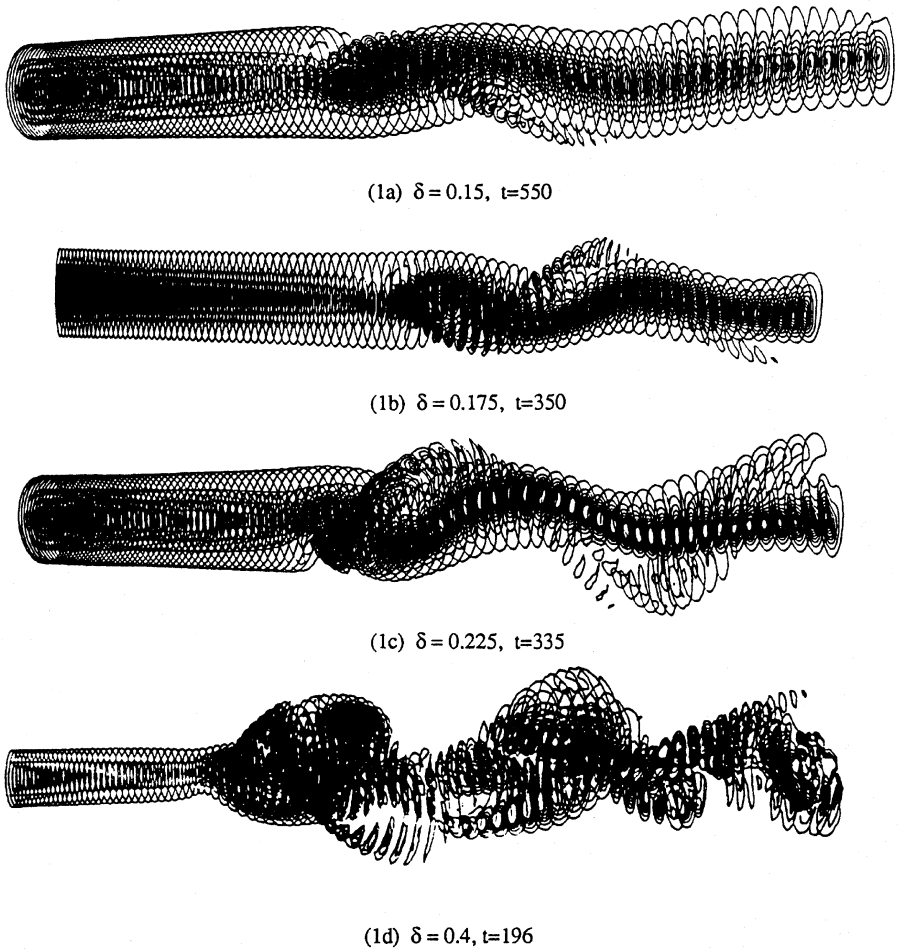


Fig. 34. Contours of normalised helicity density as the swirl level is increased [78].

As a last reference, the study of Visbal [33] of vortex breakdown above delta wings pitched at high angle of attack is worth mentioning. He obtained a striking representation of a breakdown bubble (and its spiralling tail) using an iso-surface of constant total pressure (Fig. 38). This shows a great deal of resemblance to the experimental flow visualisation of axisymmetric vortex breakdown in tubes (Fig. 5).

3.4. Overview of the numerical investigations

A first numerical approach consisted of the QCA, whose failure was an indication of vortex breakdown. This gave only limited information and solutions to the Navier–Stokes equations were targeted for a more detailed insight of the phenomenon.

Owing to the complexity of the problem and the computational constraints, and despite the three-dimensional and periodic character of the flow, the early numerical computations of vortex breakdown were restricted to incompressible, laminar, axisymmetric, steady [63,64] or unsteady [65,66,68,69] flows. These computations brought solutions that resembled the experimentally observed vortex breakdown in some aspects but did not agree with all characteristics. The discrepancies were often attributed to the assumption of axisymmetry.

Spall and Gatski [67] first proposed a fully three-dimensional, time-dependent simulation of vortex breakdown, emphasising the link between vortex breakdown and vorticity distribution: the axial component of vorticity appeared to be redistributed to the radial and azimuthal components as the stagnation point was approached and a local intensification of total vorticity was noticed in the simulated bubble. However, the breakdown was not stable and tended to move upstream in time towards the inflow boundary.

Once this problem was solved, more three-dimensional, time-dependent computations followed [74–79]. One interesting conclusion is the apparent dependence of the mode of breakdown on the imposed pressure gradient. The transformation between different forms appeared,

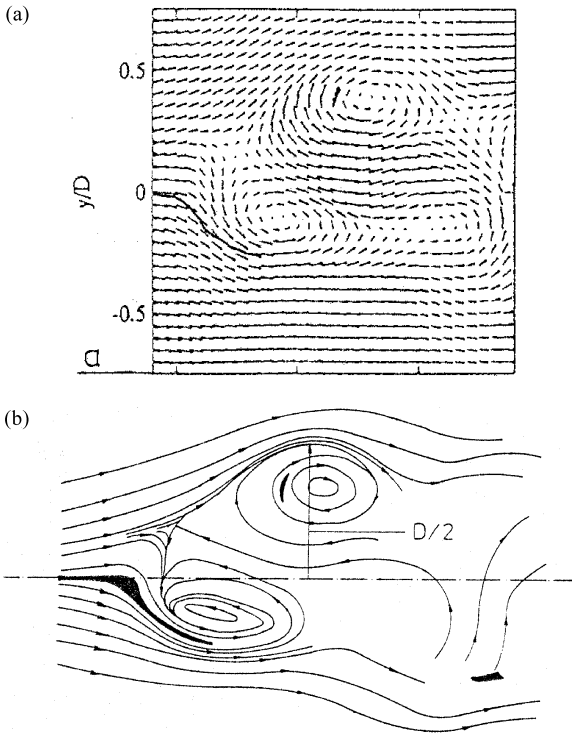


Fig. 35. Velocity vectors of a spiral breakdown obtained by PTV (a) and schematic flow pattern deduced from the PTV measurements (b) (Brüker, 1993).

however, to be gradual and not abrupt. These numerical solutions agree generally well with the experimental results. A double ring structure similar to Faler and Leibovich's [12] has been predicted, which in time, evolved towards a structure closer to that presented by Bornstein and Escudier [57].

In computed turbulent breakdown [81], the decay of axial vorticity upstream of breakdown appears to be slower than in laminar vortex breakdown. The levels of axial vorticity downstream are however much lower and much simpler (no negative vorticity).

Table 2 presents a chronological summary of the different numerical investigations examined in this section.

4. Theories and explanations for vortex breakdown

4.1. Introduction

Different theoretical goals have been pursued: earlier analyses tried to characterise the conditions under which breakdown is likely to occur, predict its location and establish criteria, while most of the recent work is aimed at describing physically the whole process and predicting the internal structure. Several theories have been proposed for vortex breakdown, which can roughly be categorised according to the following underlying ideas [16,29,30]:

1. The phenomenon is associated with the concept of critical state or, more generally, with wave phenomena [51,61,72,83,84].
2. The phenomenon is analogous to boundary layer separation or flow stagnation [42,44,60].
3. The phenomenon is a consequence of hydrodynamic instability [85,86] (cited in Refs. [16,87–89]).

Some other approaches combine several concepts and extend the previous ideas [30,55,90–93,112] or develop new arguments [32].

4.2. Wave phenomena

Independently, Squire [72] and Benjamin [51,83,84] developed theories for an axisymmetric, inviscid, and steady vortex. They both suggested the existence of a 'critical state' which separates the sub from the supercritical flow state. In a subcritical flow, disturbances can propagate upstream and downstream and standing waves are supported, whereas in a supercritical flow, only downstream propagation is possible.

Although Squire and Benjamin shared the same definition for the critical state, their interpretations, as pointed out by Hall [29], were different: Squire understood breakdown as an accumulation of upstream travelling disturbances at the locus of the critical state (like a shock wave), whereas Benjamin thought of breakdown as an abrupt change, a transition with finite amplitude, between two conjugate flow states (like a hydraulic jump in an open-channel flow). Stuart [30] highlighted the analogy between the characteristic parameters of these two different phenomena: both the Mach ($Ma = u/a$) and Froude numbers ($Fr = u/(gh)^{1/2}$) are ratios of velocity to wave velocity.

As explained by Escudier [16] in his review, the underlying idea of Squire's theory was that if standing waves can exist on a vortex flow, then disturbances, which are generally present downstream, will propagate upstream and cause breakdown. In order to quantify the level of swirl, Squire defined a characteristic parameter k, ratio of the maximum swirl velocity to the axial velocity. Applying his analysis to different swirl distributions, he concluded that there exists a critical value of k above which standing waves can be supported, the value depending on the initial swirl distribution (comprised between 1 and 1.2 for an exponential distribution and a Rankine vortex, respectively, with the axial velocity profile being constant). At the critical value, waves are of infinite wavelength. If the parameter k is assumed to increase streamwise, due to viscosity or geometry effects, an initially supercritical flow ($k < k_{crit}$) could pass through the critical state ($k = k_{crit}$) and then become subcritical ($k > k_{crit}$). Breakdown would then be expected to occur just at the critical location. Hall [29]

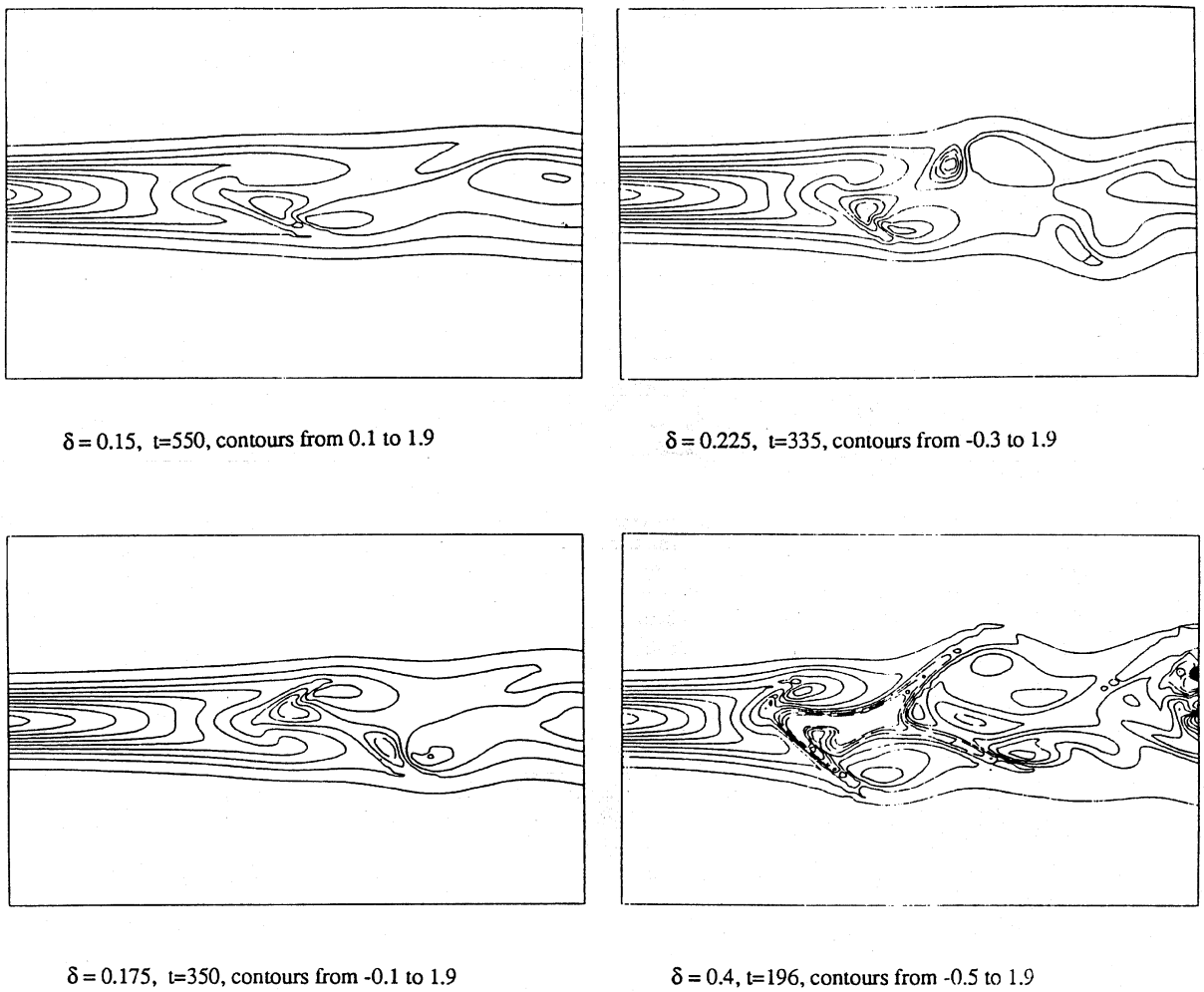


Fig. 36. Axial vorticity contours as the swirl level is increased [78].

criticised this approach as it does not suggest the abrupt change in core structure observed experimentally. Benjamin [51] also contested this interpretation, pointing out that whilst the phase velocity of the standing waves considered by Squire might be directed upstream, their group velocity was directed downstream so that no energy could be transported upstream and waves could form only in the rear of a disturbing agency. For Benjamin, the

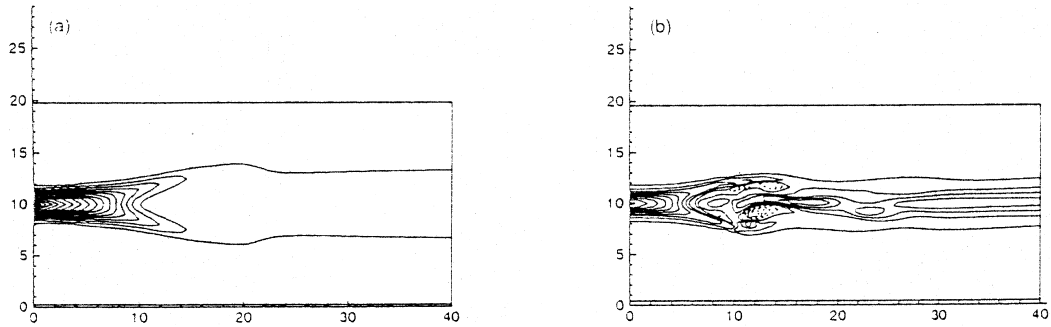


Fig. 37. Contours of axial vorticity (a) ARSM (contour levels from 0.1 to 1.9 in intervals of 0.1); (b) laminar breakdown (contour levels from −0.7 to 1.9 in intervals of 0.2) [81].

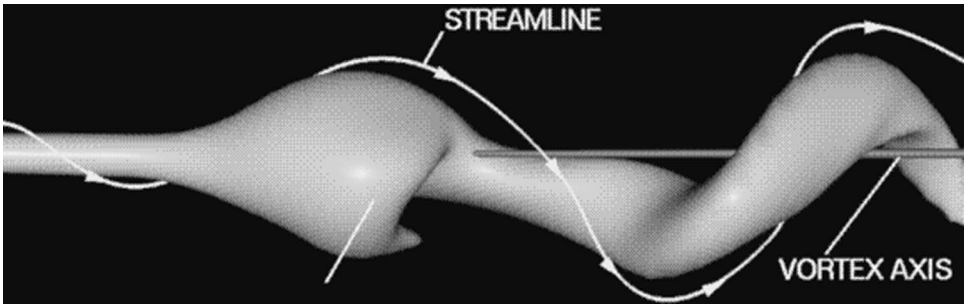


Fig. 38. Iso-surface of constant total pressure in delta-wing bubble breakdown [33].

criterion presented by Squire only indicated the critical state that indeed plays an important, though not exclusive, role in the phenomenon.

From his analysis, Benjamin [51] concluded that the subcritical state always had a greater momentum flux or flow force than the supercritical and postulated the existence of small standing waves on the subcritical flow to compensate. The argument was applied only to transitions involving small differences of flux. When the transition is nondissipative the perturbation is a solitary wave, while a small amount of dissipation would yield a wave train. He thus suggested that the breakdown is not *caused* by the existence of waves, but rather *is* the leading wave, for weak transitions. To justify strong transitions, turbulence was suggested. Referring to the observation that the core of the original flow was deformed into a spiral following the breakdown, Benjamin stated that *although the essential mechanism of vortex breakdown is explainable in terms of an axisymmetric model*, *there may be in practice considerable disturbances from this basic situation*. Although this approach is very attractive (abrupt change, occurrence and position of vortex breakdown greatly influenced by downstream conditions, a bubble resembling a wave train with the leading wave intact and succeeding waves increasingly distorted by instabilities and other disturbances, etc.), it presents several shortcomings. Experimentally, strong transitions, involving stagnation of the axial flow, large axial gradients of velocity and a pronounced divergence of the stream tubes, can be quite smooth and turbulence free. The flow immediately upstream of breakdown is required only to be supercritical, which covers a large range of possibilities, whereas Squire's explanation specifically requires the flow there to be near critical. Contrarily to Benjamin's conclusions, force and loss-free transitions have been shown to be possible by Keller et al. [93] if the conjugate flow state is generalised to include hollow-core flows.

As in Squire's work, a characteristic parameter was defined that delineates the critical regions of the flow. This parameter, N, was analogous to the Froude number for open channel flow, or Mach number for compressible flow:

$$N = \frac{C_+ + C_-}{C_+ - C_-} = \frac{u}{C} \tag{3}$$

where C is the relative speed of wave propagation; $c_+ = u + C$ the downstream propagating waves; $c_- = u - C$ the upstream propagating waves. The flow is supercritical if $N > 1$ and subcritical if $N < 1$.

Applying his theory to a combined vortex, Benjamin [51], although starting from a different perspective, arrived at the same critical condition described by Squire [72], i.e. $k \geq 1.2$ [4].

Lambourne [38] also reached similar conclusions when examining, theoretically, the effect of an imposed pressure gradient on an inviscid model vortex. The results, as well as indicating the possibility of the formation of an inner core of reversed flow, show the existence of critical conditions, depending on the ratio of the rotational and axial velocity components, beyond which drastic changes of flow structure must occur as no solution can be found. For the same flow conditions as Squire and Benjamin, without an applied pressure gradient, the same critical value was again obtained ($k \geq 1.2$). When an imposed pressure gradient was applied, the critical value was found lower, and even lower than 1. This agreed with Sarpkaya's observation [46] that breakdown in adverse pressure gradients can occur with maximum swirl velocities less than the axial.

Bossel [61] stated that 'vortex breakdown is a necessary feature of supercritical viscous vortex flow having high swirl close to the critical and some flow retardation at and near the axis, perhaps caused by an exterior pressure gradient, or by an object on the axis itself. Vortex breakdown is fully explainable and describable by supercritical solution to the inviscid equations to which the Navier–Stokes equations approximate in the breakdown region.' He divided the flow field into different regions in which he used different approximations to the Navier–Stokes equations. The occurrence of breakdown-like structures was predicted which, however, depended on the assumed condition downstream. Hall [29] stated that 'Bossel's main contribution is a demonstration of the consistency between abrupt changes in flow structures and the critical state'.

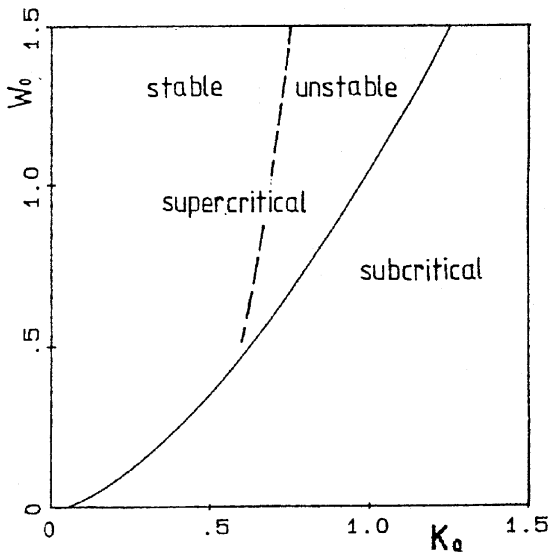


Fig. 39. Different flow states as a function of the type of flow, i.e. jet-like (axial velocity $W_o > 1$) or wake-like (axial velocity $W_o < 1$), and the swirl strength K_o.

Experimental evidence tends to support the criticality approach: all experimental studies seem to confirm that the flow upstream of vortex breakdown is supercritical, while the flow downstream of it is subcritical [94]. However, numerically, a certain controversy exists: Grabowski and Berger [64] stated that breakdown-like flows were obtained numerically from both supercritical and subcritical initial flows; and Leibovich [7] pointed out that all the initial flows considered by Kopecky and Torrance [63] were subcritical. Nevertheless, numerical solutions of the Navier–Stokes equations, reported by Shi [65], revealed breakdown structures only for supercritical approach flows. Starting from subcritical initial flows, Shi could not obtain a converged solution. Parallel calculations conducted with the quasi-cylindrical approximation brought similar conclusions [95].

Breuer [96] (cited in Ref. [31]) found a super/subcritical transition in the flow field of a steady vortex that did not display any breakdown and concluded that the transition from super- to subcritical flow state is only a necessary, but not sufficient condition for vortex breakdown.

4.3. Flow stagnation

One feature, generally accepted as an essential aspect of vortex breakdown, is the stagnation of the flow at or near the vortex axis. On the basis of this characteristic, the position and occurrence of vortex breakdown could be estimated by the quasi-cylindrical approximation (QCA). This method has already been reviewed earlier. However, it is interesting to note that [95] demonstrated that failure of the QCA and critical state are related. They showed that the critical state corresponds to a singularity. Under the QCA, if the axial and azimuthal velocity profiles are given, then the radial velocity profile satisfies a linear ordinary differential equation of second order. As a flow shifts from a supercritical state to a subcritical one by increasing the swirl level continuously, the radial velocity profiles separately tend to positive and negative infinity. Calculations with the QCA must then necessarily fail at this point. It has to be underlined that the equation giving the radial velocity profile is merely one term more than the test equation of the critical classification [51]. This is a nonhomogeneous term, related to viscosity, which causes the point of failure to precede the critical point (Fig. 39).

Under the assumption of inviscid flow, Ludwieg [97] and Hall [29] both verified that the point where calculation using the QCA fails corresponds exactly to the critical state. These inviscid calculations were criticised and judged meaningless by Shi and Shan [95] as *the only solution is the trivial one*. Krause [44], however, got some insight into the phenomenon (initiation of the process and effect of imposed pressure gradient) by comparing inviscid and viscous solutions.

For Gartshore [60], vortex breakdown was due to the diffusion of vorticity accompanying the axial flow reversal, rather than to disturbance propagation. Nevertheless, he added that the flow downstream of this reversal might be unstable. He proposed that the observed spiral flow is evidence of this instability. Grabowski and Berger [64] also stated that in the absence of an externally imposed pressure gradient or flow divergence, breakdown results from the diffusion and convection of vorticity away from the vortex core.

While this method seems to be able to predict the occurrence and position of breakdown, it is, however, limited as it cannot account for upstream influences and cannot predict the behaviour of the flow beyond the stagnation point.

4.4. Stability theory

The stability of vortex flows has been studied by numerous authors, including Rayleigh [98], Jones [99,100], Ludwieg [85,86], Howard and Gupta [87], Lessen et al. [88], Leibovich and Stewartson [89].

The first important conclusion was brought by Rayleigh, stating that a circular inviscid flow is stable if, and only if, the angular momentum increases at least monotonically outward [101]:

$$\frac{\mathrm{d}(wr)^2}{\mathrm{d}r} > 0 \tag{4}$$

In tubes, since the swirl velocity component goes to zero at the wall, an inertial instability is always present if the Reynolds number is high enough. This may result in Taylor–Gortler vortices confined to the wall region [94].

Jones [99] was the first to associate hydrodynamic instability to vortex breakdown. He showed that a Rankine vortex is unstable only to spiral disturbances and that the

Hall vortex, with constant axial velocity, is stable to axisymmetric disturbances in the limits of the Rayleigh criterion [16]. For the specific case of the Hall vortex, he was unable to draw a general criterion.

Howard and Gupta [87] proposed a generalised Rayleigh's criterion for the stability to *axisymmetric* disturbances of flows with an axial component of velocity:

$$\frac{\mathrm{d}}{\mathrm{d}r}(r^2w^2) > \frac{1}{4}r^3\left(\frac{\mathrm{d}u}{\mathrm{d}r}\right)^2 \tag{5}$$

Ludwieg [85] derived another criterion for instability to *spiral* disturbances of an inviscid swirling flow in a narrow annulus. It was a generalised form of Rayleigh's criterion for centrifugal instability:

$$(1-h)(1-h^2)-(5/3-h)g^2 < 0 \tag{6}$$

where

$$g = \frac{r}{w}\frac{\mathrm{d}u}{\mathrm{d}r}$$

$$h = \frac{r}{w}\frac{\mathrm{d}w}{\mathrm{d}r}$$

He applied his theory for helical flow in a cylindrical annulus to the Hall vortex and, eventually, extended his arguments to the vortex breakdown despite the considerable simplification allowed by the geometry and the restrictive boundary conditions imposed [89]. Sarpkaya [11] commented that this theory could not explain the axisymmetric type of breakdown and remarked that Ludwieg's test criterion predicted instability for a wide range of quasi-cylindrical flows, including some that did not exhibit vortex breakdown. Escudier [16] also pointed out the limitation of Ludwieg's analysis as it does not distinguish between modes of disturbance while vortex flows have been shown later to be much less stable to spiral disturbances than to axisymmetric disturbances.

A more general criterion has been proposed by Leibovich and Stewartson [89] for instability of an unbounded columnar vortex embedded in an inviscid flow:

$$w\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{w}{r}\right)\left[\frac{\mathrm{d}}{\mathrm{d}r}\left(\frac{w}{r}\right)\frac{\mathrm{d}}{\mathrm{d}r}(wr)+\left(\frac{\mathrm{d}u}{\mathrm{d}r}\right)^2\right] < 0 \tag{7}$$

However, when applied to a stable slender vortex [96], regions of instability towards spiral disturbances were found to exist even though no breakdown could be observed. Althaus et al. [31] suggested that the discrepancy might come from the inviscid formulation that does not consider viscous damping effects. The criterion was therefore judged not to be a sufficient condition for vortex breakdown.

A linearised inviscid stability analysis considering different modes of disturbance has been carried out by Lessen et al. [88]. They calculated the magnitude of the swirl required to stabilise jets and wakes against nonaxisymmetric disturbances on a flow with a Q vortex. The general form of the disturbance being of the type:

$$\exp[\mathrm{i}(\alpha z + m\theta - \alpha ct)] \tag{8}$$

where z and θ are, respectively, the axial and azimuthal coordinates, t the time, α the axial wavenumber, m the azimuthal wavenumber, and c is the complex phase velocity.

They found that, as the swirl is continuously increased from zero, the disturbances die out quickly for a small value of q if $m = 1$ ($q = 1.57$ ($w_{max}/\Delta u$)). However, the negative modes (i.e. helical wave paths opposite to the sense of rotation of the vortex) are initially destabilised by the swirl: the amplification rate increases and then decreases with q, falling to negative values at q slightly greater than 1.5 for $m = -1$. The amplification rate was also found to increase with m.

Garg and Leibovich [37] fitted vortex profiles, that could be transformed into a Q vortex, to their experimental data and deduced q values. Applying Howard and Gupta's criterion, they found that their flows were stable to axisymmetric disturbances at any location, both upstream and downstream of breakdown. Applying the theoretical linear stability results of Lessen et al. [88], they also concluded that all their approaching flows were stable to nonaxisymmetric disturbances, while the wake flows were unstable. Spectral peaks for flows in the wake were also found to agree reasonably well with the calculated frequency of the most unstable $m = -1$ mode, supporting their experimental observations. They were not able to conclude whether the nonaxisymmetric instabilities promote breakdown, or are by-products. Escudier et al. [91] obtained similar results.

Later, Chao et al. [56] investigated the spectral characteristics of swirling flows and the relation between downstream instability and breakdown. They noted two dominant natural frequencies in different regions of the flow, one due to the precessing vortex core and the other to the most amplified downstream azimuthal instability. They claimed that suppressing the downstream instability, by introducing an asymmetric disturbance whose frequency 'de-excited' the instability, tended to degenerate the breakdown structure to one corresponding to a flow with a lower swirl level. They concluded that *the azimuthal instabilities are not just by-products of a primary breakdown and in some way promote the breakdown*.

From all these investigations, it seems to be generally agreed that vortex flows are much more stable to axisymmetric disturbances than to asymmetric ones. This is one argument presented by Leibovich [15] against the instability explanation: it is inappropriate to describe the purely axisymmetric breakdowns which have been observed under carefully controlled conditions in tubes [8] and in closed cylindrical containers with a rotating endwall [14].

Besides, vortex flows have been found to be unstable only to $m = -1$ disturbances, which is not consistent with all experiments: in the case of spiral breakdown, the sense of winding has been observed to be the same as the rotating flow [10,17] or opposite [13,28,39].

Furthermore, the flow observed by Harvey [8] retained a well-organised character and the normal vortex flow was even restored downstream of the bubble. He argued that if the breakdown had been essentially due to the onset of some sort of instability, the perturbation would have grown unchecked and the flow would have immediately degenerated into an unsteady random motion. The reversion to the normal flow downstream of the bubble would certainly not have occurred. Harvey suggested that this behaviour was characteristic of 'critical' phenomena and not instabilities.

Stuart [30] also considered instabilities as secondary and emphasised the crucial influence of downstream conditions by remembering that Lambourne and Bryer [28] eliminated a vortex breakdown by applying suction immediately downstream of the original breakdown position.

Other arguments were advanced by Escudier [16]. First, the phenomenon of breakdown appears, experimentally, as a sudden rather than a gradual transition. Secondly, *there is no reason for the origin of the disturbances to always be of the same rotating nature*. *Much more likely would be a disturbance with a fixed origin*, *e.g. minor imperfections in the flow geometry*, *so that the instability and spiral would also be frozen with respect to the geometry rather than rotating*. Finally, a transition from a spiral form to a bubble, which has been observed under certain conditions, would be incompatible with the instability approach, except under transient conditions.

Leibovich [15] also pointed out that a vortex flow may become unstable and not undergo breakdown and that axial flow reversals can occur without any sign of hydrodynamic instability to infinitesimal disturbances.

However, Lambourne and Bryer [28] remarked that the frequency increase accompanying a flow rate increase is a strong evidence in favour of the amplification explanation.

All these remarks do not imply that instability plays no role in flows exhibiting vortex breakdown. On the contrary, the evidence seems to suggest that the wake flow is unstable. Faler and Leibovich [12] declared that *the appearance of azimuthal asymmetries in a flow with boundary conditions that are*(*experimentally*) *quite axially symmetric*, *and the important effects these asymmetrical motions have in the breakdown region show that instability is indeed manifest in these flows*. Furthermore, the absence of asymmetries in more controlled conditions, like in cylindrical containers with a rotating endwall [14], supports this conclusion. However, this concept does not seem to be the central explanation to vortex breakdown.

4.5. Combinations of different theories

As the different explanations, independently, cannot completely and accurately describe all the features of the phenomenon, and do present shortcomings, combinations have been proposed.

For example, vortex breakdown has been postulated to be the result of a process combining wave propagation and hydrodynamic instability phenomena in an intimate way.

Randall and Leibovich [90] extended the concept of Benjamin and modelled the bubble form of vortex breakdown within slowly diverging tubes, as a standing, axisymmetric trapped wave. As deceleration of the basic flow causes energy transfer from the flow to the wave, the slight divergence of the tube walls can cause the amplification of the waves travelling upstream. Wave propagation is however limited if dissipation is present and an equilibrium, resulting from the competition between amplification and dissipative effects, can be reached: the stationary wave is 'trapped'. Wave trapping has been found to be possible only if the wave had reached a particular amplitude, which also appeared to be sufficient to lead to stagnation. Their model was based on geometrical parameters and experimental data from Ref. [10]. Many of the experimentally observed features of vortex breakdown were reproduced, including the influence of an adverse pressure gradient, a well-defined closed axisymmetric cell of recirculating fluid approximately the same size and aspect ratio as that observed by Sarpkaya [10], and a wall pressure behaviour consistent with Sarpkaya's [10] and Kirkpatrick's [48] observations. Furthermore, the flow was found supercritical upstream of breakdown and subcritical downstream and the calculated position of the breakdown as a function of Reynolds number agreed well with that of Sarpkaya's [10]. One criticism against this proposal was that the model was centred on a theory of long *weakly* nonlinear waves, and could not therefore represent a logical explanation to a *strongly* nonlinear phenomenon [29]. Furthermore, as wave trapping only occurs at large amplitudes, the theory had to be extended beyond its limits of validity to capture the phenomenon. *Unreported work by Leibovich suggests*, *however*, *that wave motions of the form proposed here but of arbitrary amplitude may be possible* [94]. Leibovich [7] also highlighted a shortcoming of the model: reversed swirl velocities were predicted inside the recirculating region, which was physically impossible for a steady axisymmetric flow. Moreover, Randall and Leibovich [90] stated that reversal of tangential velocities had not been reported experimentally. The model was thus judged inadequate in describing the flow inside the breakdown. The prediction of reversed swirl velocity was thought to be due to the inviscid assumption, as in Ref. [61], and Leibovich concluded that the flow in the interior of a breakdown was then controlled by viscosity if laminar or by turbulent Reynolds's stresses if turbulent. However, reversed swirl velocities have been detected experimentally in bubble breakdown by Brücker and Althaus [59]. Another weakness of this theory concerned the dissipation of wave energy. Experimental evidence indicated that viscosity did not greatly affect vortex breakdown. Furthermore, it would not be sufficient. Additional dissipation through turbulence could not be invoked either since breakdown also occurs in a laminar vortex. Huang [102], one of Leibovich's co-workers, suggested that another

'pseudo-dissipation' was possible, if nonaxisymmetric perturbations were allowed in addition to the axisymmetric ones (i.e. if the flow is unstable to such perturbations). In that case, energy could be transferred between axisymmetric and nonaxisymmetric waves in a conservative system. This scenario integrated the axisymmetric wave required to cause stagnation and the asymmetric features experimentally observed [94]. A theoretical confirmation has not yet been produced to the author's knowledge, but a first analysis of the stability of a columnar vortex flow [94] seems to support the idea that instability to nonaxisymmetric disturbances and flow retardation are related.

Bilanin [103] (cited in Ref. [16] also supported the idea that wave trapping is related to vortex breakdown. Contrary to Benjamin, he considered the criterion to be based on the group-velocity going to zero rather than the phase velocity. Tsai and Widnall [104] also investigated a group-velocity criterion on real vortex flows that contain breakdown. As different wave modes propagate with different group velocities, a flow may be supercritical to axisymmetric waves while, at the same location, be subcritical to asymmetric waves. Using the same approach as Garg and Leibovich [37], they fitted velocity profiles to their experimental data and used a parameter q for stability conditions. They reached conclusions similar to these authors: for all the inlet profiles, the flow is expected to be stable to all disturbance modes ahead of the breakdown region. Downstream of vortex breakdown the flow is unstable to asymmetric disturbances. They calculated from experimental data the group velocities for both symmetric ($m = 0$) and nonsymmetric ($m = \pm 1$) modes of wave propagation. For all the flows studied, they concluded that:

- the group velocity of the $m = -1$ mode was always directed downstream;
- the flows that exhibit axisymmetric breakdown are supercritical upstream and subcritical downstream to both $m = 0$ and $m = 1$ modes;
- the flows that exhibit vortex breakdown of the spiral form are supercritical upstream and subcritical downstream only to mode $m = 0$.

The calculated frequencies of the unstable disturbances were also found to be in good agreement with the frequencies obtained from the experimental measurements.

Ma and Leibovich [105] proposed an extension of the finite amplitude wave theory used to model the bubble breakdown [90] to describe the spiral form of breakdown. They emphasised the need for both axisymmetric and nonaxisymmetric modes whichever type of breakdown is to be modelled.

Escudier et al. [91] presented evidence to suggest a connection between the vortex breakdown and criticality and stability of the vortex core upon which it occurs. Their results showed that the stability criterion [88] does not predict the location of the breakdown even though instability waves may exist. The explanation lies in the difference in the growth rate of instability waves on different types of axial velocity profiles (slow on a high-speed axial flow and rapid on a low-velocity wake-like profile). So, it is the state of criticality of the flow that determines the basic character of the downstream flow and that the waves are a superimposed, but important, fine details. They concluded that *a consequence of breakdown is a wake-like velocity profile downstream that is unstable to a discrete spectrum of spiral disturbances, and it is these disturbances that are responsible for the wide range of observed breakdown types*. This approach could explain many of the observed features. Indeed, starting from a low swirl, the core is unstable to spiral disturbances (Lessen et al.'s [88] stability criterion). In the region of jet-like axial velocity profiles, the growth rate is very slow, while it is larger in the wake-like profiles' region, which produces a mild disturbance. As the swirl is increased (or the Reynolds number, or the adverse pressure gradient, or the divergence), the back-flow becomes larger and the axial velocity lower, so that the growth of the instability takes place over a shorter distance. The breakdown region also moves upstream. This could justify the sense of winding of the spiral, the progression of forms from a mild disturbance of the vortex core to a spiral, more and more tight. At a higher given swirl, the stability criterion is reached and the vortex core is not unstable anymore: only the bubble is observed. Escudier et al. [91] noted that *in many cases the spatial growth of the instability is so rapid that a distinct axisymmetric breakdown does not have time to form before the waves dominate the observations*. They were of the opinion that *this is one of the phenomena that has confused efforts to separate the criticality and instability criteria*.

Maxworthy et al. [106] demonstrated experimentally that vortex breakdown can be caused by axisymmetric waves and possibly by combined, axisymmetric and so called 'kink' waves. Kink waves, or as described by Maxworthy et al. [107] *nonaxisymmetric kinks formed in a perturbation*, alone remain stable and do not cause disruption of the core. The possible effects of axisymmetric waves are clearly exposed. The propagation of these waves along a vortex core causes a periodic swelling and contraction of the vortex tubes. Fig. 40 shows the perturbations induced by a passing solitary wave. When the wave travels in the sense of the axial flow in the vortex (Fig. 40a), it tends to increase locally the velocity hence increasing the azimuthal vorticity or radial shear stress. On the other hand, the wave propagating against the flow induces a wake-like velocity profile and may even lead to flow reversal (Fig. 40b).

The authors suggested that the waves only propagate undisturbed up to a certain amplitude, above which the internal flow becomes unstable to spiral disturbances of negative wavenumber [88] and the vortex can become quite disturbed and turbulent (Fig. 41).

They pointed out, however, that the instability studies used are strictly valid for steady flow conditions,

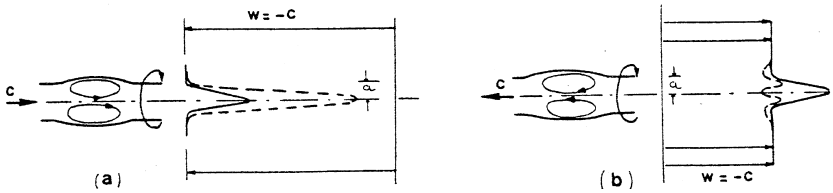


Fig. 40. Schematic of the perturbation induced by a passing solitary wave.

not travelling waves. Nevertheless, they observed experimentally that the disturbances *invariably grow to become unstable finite amplitude spirals of small pitch rotating in a direction opposite to that of the basic rotation*. This approach could allow for Sarpkaya's observation of multiple breakdown and progressive replacement of the downstream bubbles by the spiral tail as the swirl is suddenly increased (cf. Fig. 7).

The development of a fully nonlinear analysis, as already signalled by Leibovich [108], would be necessary to further progress in this direction.

An approach combining and extending Benjamin's conjugate-state transition ideas and the separation hypothesis has been presented by Escudier and Keller [55,92,112] and Keller et al. [93]. They introduced the concept of a two-stage transition, i.e. the breakdown is assumed to occur in two different stages, as illustrated in Fig. 42.

The first stage is an isentropic transition from the upstream supercritical flow to an intermediate supercritical flow, which since there is no change to a subcritical flow involves no change in the flow force (or monentum flux Benjamin [51] even for a large-scale transition. The second stage is a jump-like transition to the final subcritical flow state. It has no bearing on the breakdown criterion [16]. The intermediate supercritical state consists of a zone of nearly stagnant fluid surrounded by a region of potential flow. The two zones are separated by a layer of rotational fluid originating in the upstream vortex core, which is subject to an intense shear stress. Instability and subsequent roll-up of this shear layer, at large Reynolds numbers, was regarded

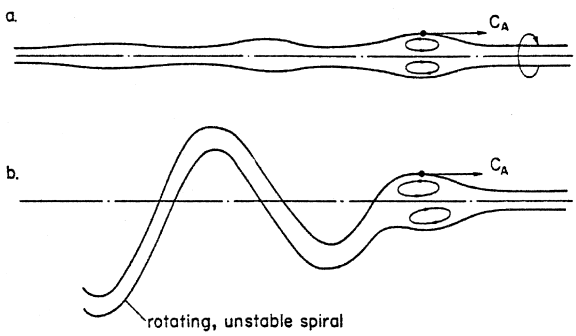


Fig. 41. (a) Axisymmetric solitary waves ordered by amplitude; (b) when the amplitude of the leading wave is large enough, the internal flow becomes unstable to spiral disturbances [107].

by Escudier and Keller [55,92,112] as the probable explanation for the occurrence of the spiral form of breakdown, whilst the basic breakdown process is axisymmetric. It was argued that the intermediate state might be masked most of the time by the shear layer behaviour (spreading at very low *Re* and instability and roll up at higher *Re*).

Different arguments were presented to support this approach. First, experimental observations (Fig. 43) showed that the fluid entering the bubble clearly originates from a region much smaller in radius than that of the bubble itself and even the core. Therefore the bubble must be a zone of essentially stagnant fluid, which has been confirmed by LDA measurements. On the same figure, the smooth appearance of the bubble surface can be observed, even when the interior is turbulent. This reinforces the idea that the first transition involves little dissipation. They also concluded that the free breakdown seems to represent another type of transition, a direct supercritical–subcritical transition, which necessarily involves dissipation, and therefore, that investigations of vortex breakdown in tubes were not directly representative of free-vortex breakdown.

Later, Keller et al. [93] suggested, in analogy to the theory of gravity currents in closed channels, the existence of two types of force-free transitions between axisymmetric flow states in tubes, one leading directly to subcritical vortices and involving dissipation, and the other occurring in two stages. They stated that for certain downstream boundary conditions, both types of transitions could exist (Fig. 44).

Escudier and Keller [92] experimented on hollow cores (air injected into the interior of a breakdown bubble in the case of a liquid flow) in order to demonstrate that such a model of breakdown was physically realistic. The viscous stress and instability mechanism were removed by this method, and the two transitions appeared completely separated, as assumed in the theoretical analysis. The smooth character of the first transition was again evident as was the more disturbed nature of the second (the hydraulic jump). As the air escaped, the two transitions moved together, ultimately coalesced, and the usual breakdown structure was recovered.

In his critical review, Stuart [30] concluded that the dependence of vortex breakdown on downstream conditions was suggestive of a relationship to criticality and that a tendency towards stagnation seemed to be at least a concomitant of the concept of criticality. He inferred this last opinion from the fact that a pressure gradient is naturally enhanced on the axis if the flow diverges, as previously

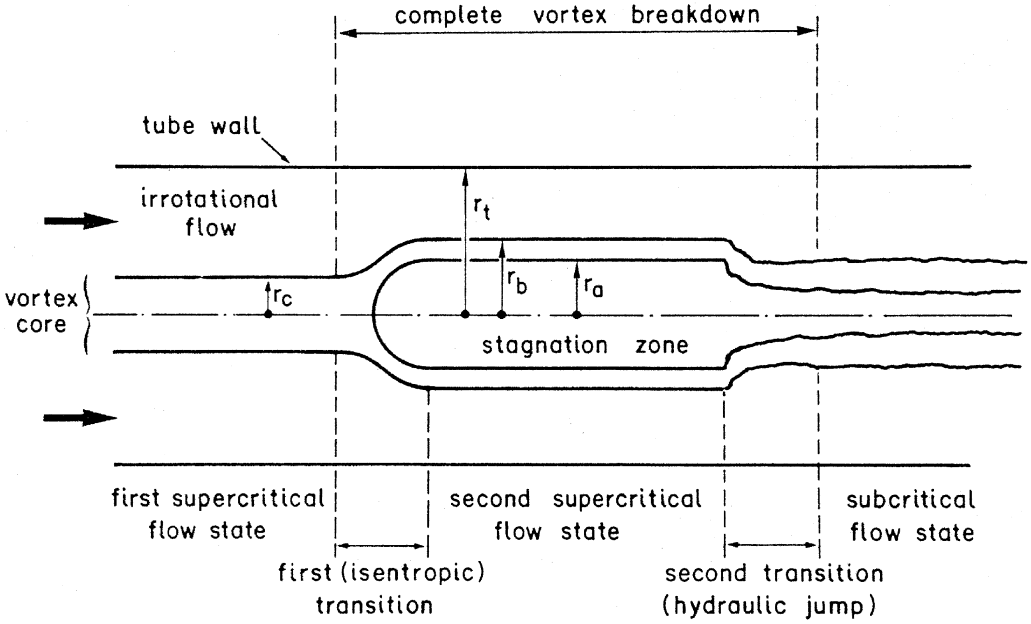
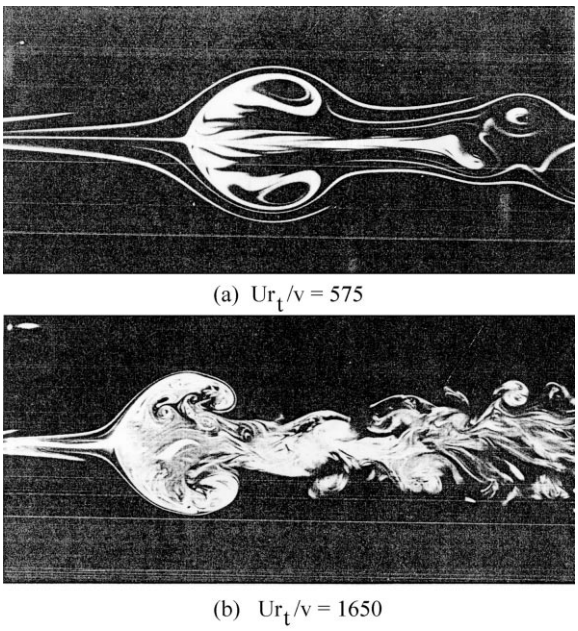


Fig. 42. Schematic diagram of the two-stages transition theory [92].

mentioned in this paper. He considered the instability effects to be secondary.

4.6. Other approaches

Escudier and Zehnder [13] proposed a simple criterion for the occurrence of vortex breakdown at a fixed location in a tube: $Re_B \sim \Omega^{-3}R^{-1}$, where Re_B is the pipe Reynolds number at which vortex breakdown occurs, Ω is the circulation number and R the ratio of the radial to tangential velocities in the inflow region. The correlation is good over a wide range of Re (5×10^2–10^5) although departures are evident for very high circulation numbers (Fig. 45). It has also to be remarked that only small values of the parameter R were employed due to the experimental arrangement.



(a) $Ur_t/\nu = 575$

(b) $Ur_t/\nu = 1650$

Fig. 43. Vortex breakdown in tubes (from Ref. [92]).

Spall et al. [4] proposed a criterion for the onset of vortex breakdown, based upon previous experimental, theoretical, and numerical studies, including their own three-dimensional unsteady simulation. The criticality condition is expressed in terms of a Rossby number Ro (or inverse swirl ratio):

$$Ro = \frac{u}{r\Omega} \tag{9}$$

where u represents the characteristic axial velocity, r^* a characteristic viscous length-scale (vortex core radius), and Ω a characteristic vortex core rotation rate.

This choice was justified by the fact that both Squire [72] and Benjamin [51] studies could be reinterpreted using this nondimensional parameter. A plot of the Rossby number Ro versus Reynolds number Re_{core} for a variety of numerical and experimental studies of swirling flows was presented (Fig. 46). These results indicated that experimentally, analytically, and computationally, Ro must be lower than a critical value, at a given Re_{core}, in order to provoke breakdown. The critical Ro appears to be approximately 0.65 for Re_{core} greater than 100. For lower Re_{core}, the value of the

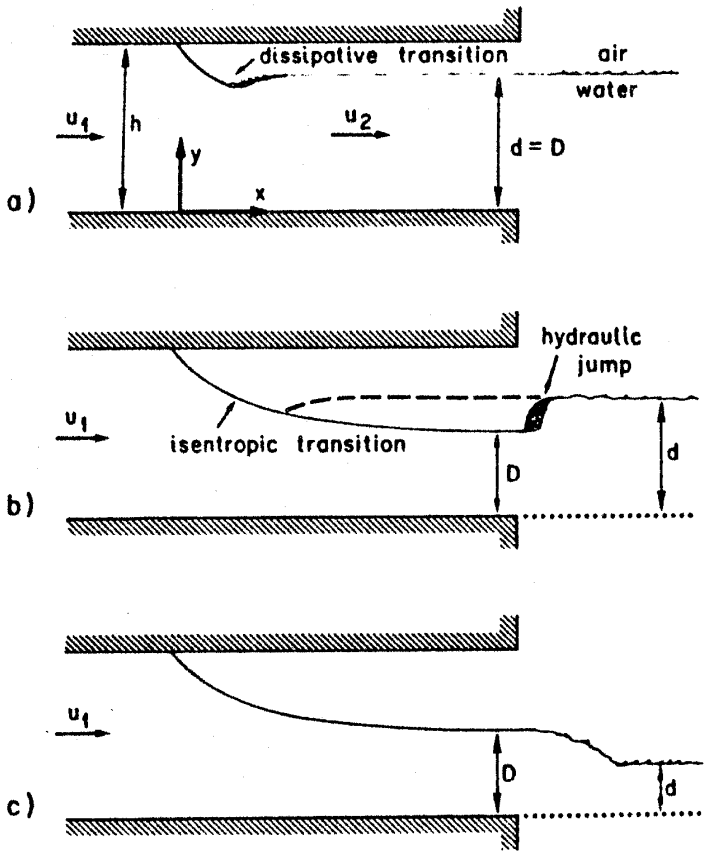


Fig. 44. Analogy to gravity currents in closed channels [93] (a) single transition to subcritical (dissipative) (b) depth *d* for which 2 solutions are possible (single or two stages) (c) small depth d for which the first transition is always is entropic.

critical *Ro* is lowered, *undoubtedly owing to the increased damping effects of viscosity on the wave motions*. The authors emphasised the fact that the established criterion can be applied to both confined and unconfined vortical flows due to the choice of characteristic scales associated with the vortex itself.

While numerous postulated criteria for vortex breakdown are based on the ratio of an axial to an azimuthal component of velocity [4,51,72], Brown and Lopez [32] developed a criterion based on the relationship between the angle of the velocity vector *and* the vorticity vector on stream surfaces upstream of this region. They proposed to explain the physical mechanisms for vortex breakdown by the production of a negative azimuthal component of vorticity. They did not find evidence of a disturbance propagating upstream and being responsible for breakdown. This study started from Escudier's [14] observations of the flow in a closed container with a rotating endwall. These revealed axisymmetric and steady recirculation bubbles over a large range of the governing parameters. Under the assumption of axial symmetry, Lopez [71] faithfully reproduced the observed flow characteristics reported by Escudier [14] and Brown and Lopez [32] gave an explanation of the vortex breakdown based on the behaviour of the azimuthal component of vorticity. Squire [72] had already used an equation for the azimuthal vorticity and Garg and Leibovich [37] had already advanced a necessary condition for breakdown (ratio of axial to azimuthal vorticity near unity). Leibovich [108] had also stated that vortex breakdown was accompanied by a change in sign of the azimuthal vorticity. However, the particularity of this study was that azimuthal vorticity was explored directly and not through perturbations of an initial stream. Brown and Lopez [32] presented an extremely simple expression between azimuthal vorticity and radius of the stream surface (σ) for a steady, inviscid and axisymmetric swirling flow:

$$\eta(x) = \frac{A}{\sigma} - B\sigma \tag{10}$$

The ratio of the azimuthal vorticity on a stream surface to its value on the surface at a location of reference (Fig. 47) was given by:

$$\frac{\eta}{\eta_0} = \frac{\sigma_0}{\sigma}\left(\frac{\alpha_0}{\beta_0}\right) - \frac{\sigma}{\sigma_0}\left(\frac{\alpha_0}{\beta_0} - 1\right) \tag{11}$$

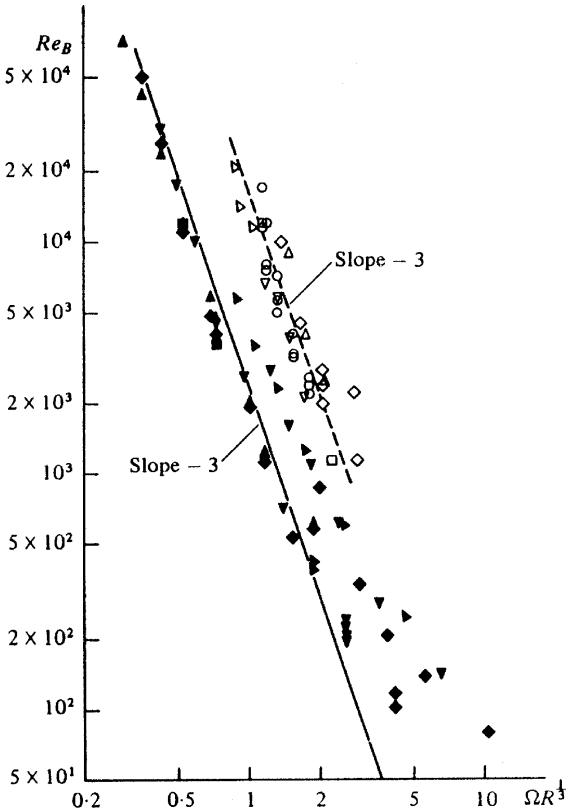


Fig. 45. Vortex breakdown data illustrating Escudier and Zehnder's criterion [13].

where $\alpha_0 = w_0/u_0$ and $\beta_0 = \eta_0/\xi_0$ (η = azimuthal vorticity and ξ = axial vorticity)

This equation is plotted in Fig. 48 for various values of α_0/β_0.

The authors also showed that, in the absence of bluff bodies in the flow field, 'the development of a negative azimuthal component of vorticity on some stream surfaces is necessary if the axial velocity is to be brought to zero.' For η_0 positive, Eq. (11) shows that η will only become negative on a diverging stream surface if $\alpha_0 > \beta_0$, i.e. $w_0/u_0 > \eta_0/\xi_0$. A helix angle for the velocity that exceeds that of vorticity on some stream surfaces seems therefore to be a necessary condition for vortex breakdown to occur. The following mechanisms were proposed. An initial divergence of the core generates a centrifugal wave and thus a waviness of the stream surface. Batchelor [109] described this process as being due to the 'elasticity' imparted to swirling flows by the Coriolis forces. The development of one or more recirculation zones is a consequence of the generation of negative azimuthal vorticity through the stretching and tilting of vortex lines. When *Re* is increased, the wavelength is reduced and the wave amplitude increased. This was judged to be consistent with the behaviour of the recirculation bubbles: as *Re* is increased, the bubble diameter increases, the leading bubble migrates towards the endwall and the distance between two bubbles is progressively reduced. A positive feedback mechanism explained qualitatively the rapid divergence of the stream surfaces in the breakdown region. 'The development of negative η on stream surfaces will induce a negative axial velocity on the axis which, by continuity, will lead to a further increase in σ and correspondingly, a further increase in negative vorticity, etc. A reduction in azimuthal vorticity with increasing radius is readily understood physically for the simple case of a flow that is cylindrical upstream and has $\eta = 0$. If two material points on a vortex line (which is axial in this cylindrical flow) move with the fluid and the leading point advances to a location where the radius has begun to increase, then this leading point will experience a reduction in azimuthal velocity due to the conservation of Γ on the stream surface.

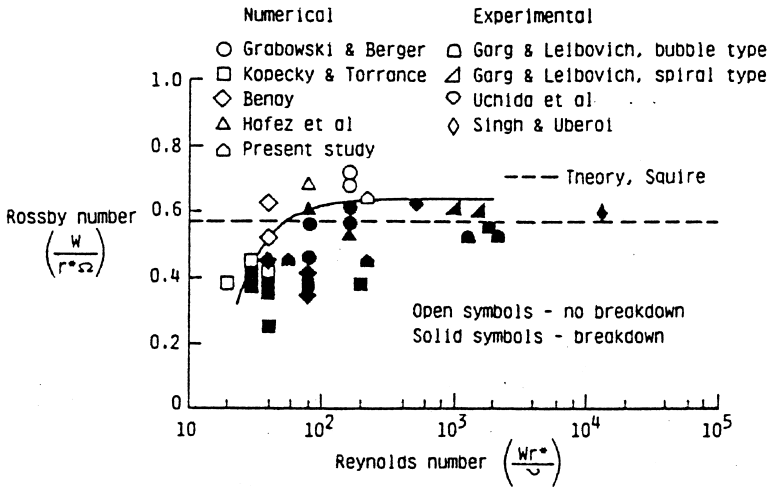


Fig. 46. Rossby number–Reynolds number relationship for vortex breakdown [4].

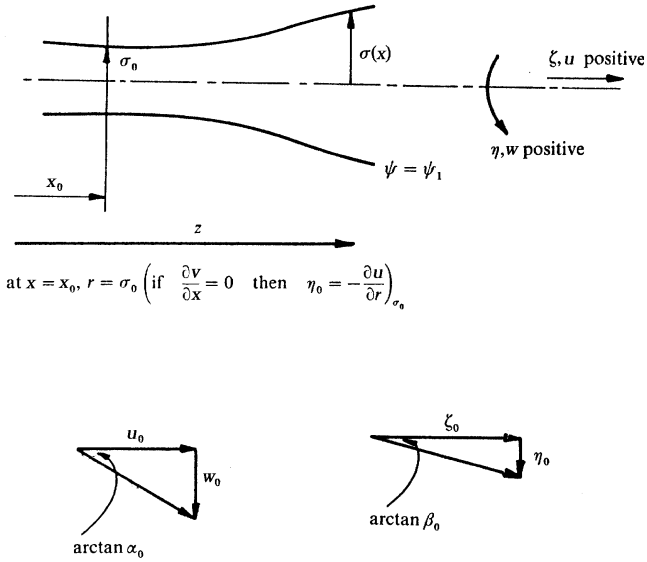


Fig. 47. Coordinate system and variables definition [32].

The material line between the points is therefore stretched and tilted, and since the vorticity moves with this material line, a negative azimuthal component of vorticity is generated. The next step was to explain why a strong vortical core flow begins to diverge. In closed containers, with a rotating endwall (their initial study), it is considered as the result of the upstream radial distribution of total head and circulation. In straight pipe flows, viscous diffusion was advanced as a mechanism of initiation of this divergence. In a (slightly) diverging duct or with the presence of a sudden expansion, the core divergence is obviously promoted by the adverse pressure gradient. This is coherent with Krause's [44] computational results for inviscid and viscous flows. He showed that for a viscous flow an externally imposed adverse pressure gradient was not needed in order to initiate the breakdown, whereas it was for an inviscid flow.

Althaus et al. [31] extended the model of Brown and Lopez [32] as they proposed, in addition to the positive feedback mechanism, a counteracting mechanism for attaining an equilibrium state (Fig. 49).

An adverse pressure gradient at large radial distances

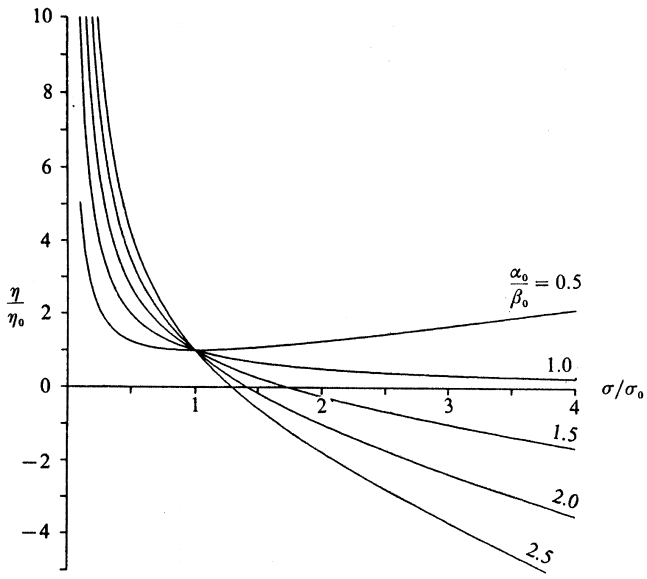


Fig. 48. Plot of Eq. (11) for different ratios α_0/β_0 [32].

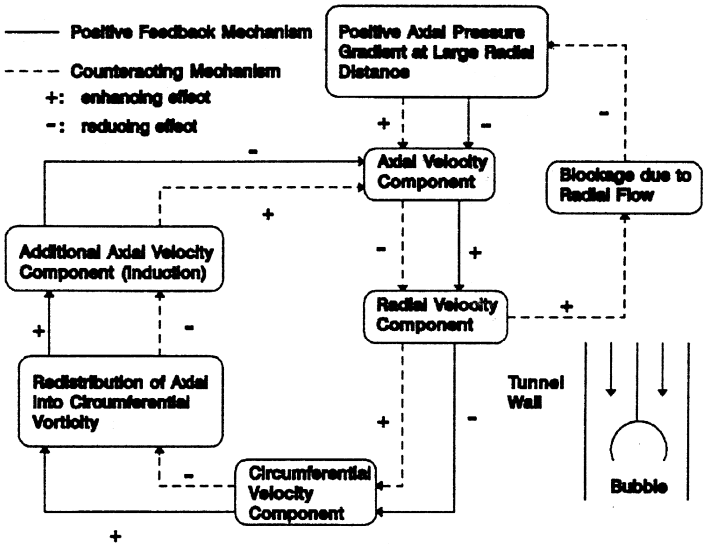


Fig. 49. Positive and negative feedback mechanisms for the bubble breakdown [31].

from the axis causes axial velocity deceleration. A radial outflow follows (mass conservation), and a redistribution of the axial vorticity into the circumferential component ensues (conservation of angular momentum and vorticity transport equation). The circumferential vorticity induces negative axial velocity and thus reinforces the deceleration on the axis. This process may lead to the formation of a stagnation point on the axis, followed by a recirculation zone. However, positive feedback alone is insufficient to explain the vortex breakdown bubble. A counteracting mechanism is required to reach an equilibrium. The radial outflow causes an acceleration of the axial velocity component at large radial distances, which reduces the effect of the adverse pressure gradient (axial deceleration), thus reducing the radial outflow. The circumferential vorticity is thus retransferred to the axial vorticity component, leading to an attenuation of the vortex ring-like structure. This process diminishes the induction of the negative axial velocity. If these positive and negative feedback mechanisms balance each other, a dynamical equilibrium state is attained with a steady position of the bubble.

4.7. Bubble and spiral breakdown: same or different phenomena?

Leibovich [15] stated that *at least two pieces of evidence exist to show that the two forms are in fact distinct flow disturbances*. The first one is the different core expansion. The diameter of the core downstream is larger than that upstream of breakdown for all forms of breakdown, but is larger for the bubble form than it is for the spiral form. Apparently, the bubble form of breakdown creates a wake that is more unstable, with greater mixing resulting in a larger core. Secondly, the transition from one form to another is associated with a discontinuity in the mean position (cf. Fig. 10 *and this discontinuity is the clearest evidence that the two disturbance forms are truly distinct*.

On the other hand, Lambourne [38] argued that the breakdown is initially axisymmetric but changes into the spiral form due to instabilities. Similarly, Escudier and Keller [55,92] considered the two forms as two different aspects of one single phenomenon which is inherently axisymmetric. They suggested that departures from axial symmetry resulted from azimuthal instabilities not directly associated with the breakdown process and leading to roll up and detachment of the shear layer surrounding a bubble. This detachment would then result in a precessive motion of the near-stagnant interior fluid and in a spiral appearance. The experimental results of Escudier and Zehnder [13] tend to support this idea of a unique phenomenon as the data correlation established is independent of the breakdown's appearance. A further point put forward by Escudier and Keller [92] was that if the spiral form was inherently nonaxisymmetric, then the structure would have to remain stationary instead of rotating with the fluid, as it was observed for the nonaxisymmetric standing-wave patterns in hollow-core vortices [110].

According to Lowson [41] and Jones [100], however, the phenomenon, attendant to leading-edge vortices, always commences as a spiralling of the axial filament while the axisymmetric form is *a late development of the primary spiral form*. Similarly, in a slightly diverging duct, Sarpkaya [11] suggested that the spiral is the *result of an instability in flows with moderate swirl*. Hall [29] suggested that these instabilities might not occur in flows with high swirl, which would then result in an axisymmetric bubble. This is coherent with Lessen et al.'s [88] work showing that the negative modes of disturbances are only initially destabilised

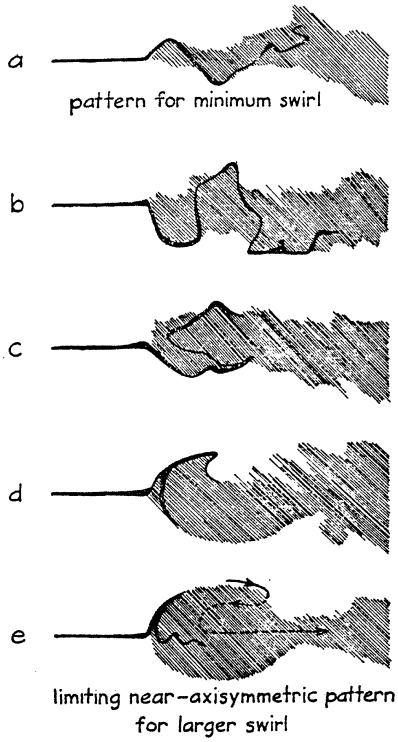


Fig. 50. Evolution of breakdown pattern with swirl increase [29].

by the swirl. On the basis of Sarpkaya's [10] observations Hall [29] presented sketches describing the variation of the breakdown pattern with increase of swirl (Fig. 50). As the vane angle is progressively increased, the dye filament is distorted and tends to turn back towards the kink.

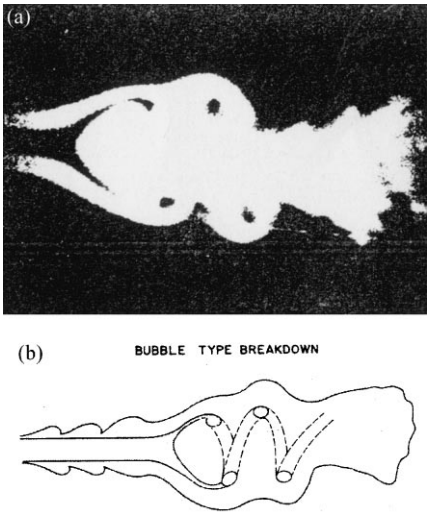


Fig. 51. Visualisation and schematic representation of a bubble breakdown over a delta wing [36].

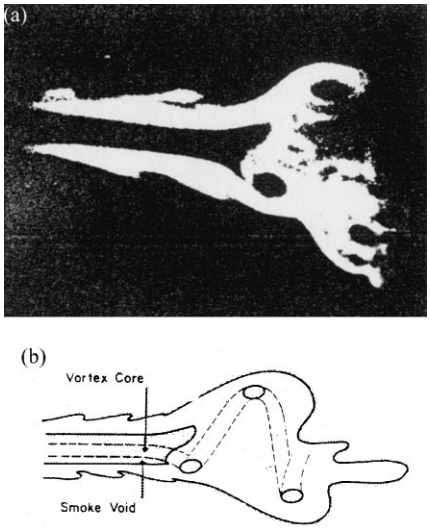


Fig. 52. Visualisation and schematic representation of a spiral breakdown over a delta wing [36].

Eventually, a bubble-like structure emerges. It is not closed at the rear. Hall's sketches resemble very much the structure proposed by Uchida et al. [58] (cf. Fig. 19).

A similar interpretation has been suggested by Payne et al. [36], in a study of vortex breakdown over a delta wing. The core in a bubble form of breakdown (Fig. 51a) seems to expand around an oval-shaped recirculation zone, at the exit of which it appears to shed in the form of vortex rings that are then convected downstream. The authors believed that these 'vortex rings' were actually a consequence of a tightly wound spiral tail originating from the downstream end of the bubble (Fig. 51b).

The authors also observed an occasional change of structure into what might be interpreted as a spiral mode (Fig. 52). This transformation was not instantaneous, but rather a continuous process, and was accompanied by a downstream movement of the breakdown's mean location. After a short time, the spiral transformed back into the 'bubble' and moved upstream.

As mentioned previously, Brücker [39] also concluded from PTV measurements that bubble and spiral breakdown seemed to be very similar. He stated that the different observable forms seemed to be determined by the slope, winding and diameter of the spiral vortex-core. It was argued that the deflection of the vortex core near the breakdown point induced an asymmetric reversed flow, whose strength and position depended on the slope of the deflected vortex core. This process was responsible for the radial distance between stagnation point and centreline (Fig. 21). The bubble breakdown was assumed to be a compressed spiral in which the induced backflow was more axisymmetric with a stable stagnation point at the centreline. In that case, the fluorescent dye filament was no longer deflected as a whole but spread away and filled up the layer between the outer flow

Table 3
Chronological list of some theortical investigations on vortex breakdown

Year	Author(s)	Type of theory	Main aspects
1960, 1964	Jones	Hydrodynamic instability	Stability to axisymmetric and spiral disturbances
1960	Squire	Criticality	Accumulation of upstream travelling disturbances — criterion ($f(w/u)$)
1962, 1965, 1967	Brooke Benjamin	Criticality	Transition between conjugate flow states — criterion ($f(w/u)$)
1962, 1963	Gartshore	Stagnation — separation analogy	Stagnation due to vorticity diffusion — spiral due to instability of the flow downstream of the reversed flow region
1962	Howard and Gupta	Stability	Generalised Rayleigh's criterion for the stability to axisymmetric disturbances
1962, 1965	Ludwieg	Stability	Stability to spiral disturbances
1966, 1967	Hall	Separation analogy	Failure of the q_{ca}
1967, 1969	Bossel	Criticality — flow retardation	Various approximations of Navier–Stokes equations in different regions of the flowfield
1967	Lambourne	Criticality — effect of an imposed Pgrad on an inviscid model vortex	Possibility of formation of an inner core of reversed flow — existence of critical conditions ($f(w/u)$)
1972	Hall		Review and criticism
1973	Bilanin	Criticality	Trapped wave — criterion based on group-velocity
1973	Randall and Leibovich	Criticality — weakly nonlinear wave propagation	Bubble modelled as a standing, axisymmetric trapped wave
1974	Huang	Criticality — weakly nonlinear wave propagation	Trapped wave — energy transfer between axisymmetric and nonaxisymmetric waves
1974	Lessen, Singh and Paillet	Stability	Linear inviscid stability analysis
1980	Keller and Escudier	Hollow core vortices	Effect of axisymmetric (contraction) and asymmetric (elbow) disturbances — stationary counter-helix with the elbow
1980	Tsai and Widnall	Criticality	Group-velocity criterion tested on real vortex flows containing breakdown — different criteria for different wave modes (different group velocities)
1982	Escudier and Zehnder		Criterion (Re[a], Ω[b], R[c])
1982	Escudier, Bornstein and Maxworthy	Criticality — stability	Breakdown due to criticality — wake-like velocity profile downstream may be unstable to spiral disturbances
1983, 1985	Escudier and Keller	Criticality — two-stages transition	Transition structure — spiral form of breakdown due to azimuthal instability leading to roll up and detachment of the shear layer surrounding the bubble — criterion
1983	Leibovich and Stewartson	Stability	General criterion for stability of an unbounded columnar vortex embedded in an inviscid flow
1985, 1987	Keller et al.	Criticality — two-stages transition	Transition structure — criterion
1985	Ma and Leibovich	Criticality	Extension of Randall and Leibovich's (1973) wave theory to the spiral form
1985	Maxworthy et al.	Wave propagation — stability	Vortex breakdown can be caused by axisymmetric waves, and possibly combined axisymmetric and kink waves
1987	Shi and Shan	Criticality — separation analogy	Failure of the QCA and critical state are related
1987	Spall, Gatski and Grosch		Criterion based on Rossby number
1987	Stuart	Review criticality — stagnation	Review stability secondary
1988	Escudier		Review
1990	Brown and Lopez	Production of a negative azimuthal component of vorticity — stagnation	Bubble breakdown — criterion for a reversed flow region based on the relationship between the angle of the velocity vector *and* the vorticity vector on stream surfaces upstream of this region — positive feedback mechanism
1995	Althaus et al.	Vorticity redistribution — stagnation	Bubble breakdown — positive and negative feedback mechanisms

[a] Re based on the tube diameter.
[b] Ω = circulation.
[c] $R = v/w$ in the inflow region.

and induced reversed flow suggesting a bubble shape. Due to the compression, the azimuthal vorticity was concentrated in a vortex-ring like structure and, experimentally, it was no longer possible to observe clearly separated vortices. Through this approach, Brücker also explained the upstream movement of the vortex breakdown observed experimentally when changing from a spiral form of breakdown to a bubble: due to the compression, the total vorticity is more concentrated in azimuthal direction. The induced backflow is therefore stronger, which changes the forces providing equilibrium and causes an upstream movement. This proposition is extremely attractive as it justifies many features, such as the sense of winding of the spiral opposite to the outer flow, the asymmetric and periodic flow in the bubble, the toroidal ring observed at the downstream end of the bubble, the definite order of progression of the different forms as Reynolds number and swirl are increased, the relative position of bubble and spiral, etc. However, multiple breakdowns and downstream influence, observed experimentally, cannot be explained by this sole approach. Nevertheless, the author believes that Brücker's [39] observations can be somehow related to Escudier et al.'s [91] conclusions suggesting that the breakdown causes a wake-like profile downstream that is unstable to spiral disturbances.

4.8. Overview of the theoretical investigations

A chronological table listing the publications used in this study and related to theories and explanations of vortex breakdown is presented in Table 3.

Many interesting interpretations have been brought up by different investigators. However, while an explanation appears to agree very well with one (or several) characteristic(s) of the phenomenon, it usually cannot satisfy all the observed features. Many explanations also complement each other. Criticality seems to be an essential part of the vortex breakdown process. Indeed, downstream conditions do appear to be crucial in vortex breakdown, and all the flows were observed to be supercritical upstream of breakdown and subcritical downstream. Stagnation is commonly accepted as a characteristic of the phenomenon. Instability also exists. In several studies, these different notions have been intimately linked together. Shi and Shan [95] showed that stagnation and critical state are related. Randall and Leibovich [90] presented the concept of wave trapping, combining wave propagation and hydrodynamic instability. Similarly, Escudier et al. [91] and Maxworthy et al. [106,107] suggested a connection between vortex breakdown and criticality and stability of the vortex core upon which it occurs, stability being a superimposed phenomenon. Brown and Lopez [32] also considered stability as secondary and disregarded the role of criticality. Instead, they proposed a mechanism, based on the production of a negative azimuthal component of vorticity. This study was limited to the breakdown bubble. Brücker's [39] proposition (different forms determined by the slope, winding and diameter of the spiral vortex core) brings an interesting view on the phenomenon.

Considering all these studies, it appears that a complete scenario is still to be proposed and approved. However, a clearer structure of the flow pattern has emerged.

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